Housing Affordability Stress in Canada: A Household-Level, Data- Driven Support System

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Field Project

Submitted in partial fulfillment of the requirements for the

Master of Data Analytics

Graduate Studies

University of Niagara Falls

Niagara Fall, ON

March 2026

Signature Page

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Abstract or Executive Summary

Housing affordability stress continues to pose a significant policy concern in Canada; nevertheless, current methodologies frequently depend on aggregate measures that may not adequately reflect disparities among households. This constrains governments' capacity to precisely identify at-risk groups and deliver scarce housing assistance efficiently. This work fills this gap by creating a household-level, data-driven decision support system to identify and prioritize housing affordability stress.

The analysis employs the 2022 Canadian Housing Survey (CHS) Public Use Microdata File, utilizing households as the unit of analysis. Core Housing Need (PCHN) functions as the principal outcome metric. A systematic analytics methodology was employed, including weighted descriptive analysis, interpretable predictive modeling, and prescriptive decision-support strategies. A logistic regression model was created to assess the likelihood of housing stress, using Random Forest as a reference, and the findings were implemented via risk segmentation, threshold optimization, and a policy tier structure.

The results demonstrate a significant correlation between housing affordability stress and parameters such as tenure, income, and geographical characteristics. Renters have significantly elevated stress levels compared to homeowners, with low-income households representing the greatest concentration of need. Weighted estimates indicate that roughly 11.39% of Canadian households endure housing stress, but the predictive model identifies 10.52% of homes as high risk at a policy-relevant threshold. The model exhibits robust screening efficacy (ROC-AUC = 0.9188), attaining a recall of 0.80 at the designated threshold, signifying its proficiency in detecting a substantial proportion of at-risk families.

The analysis indicates that housing stress exists on a continuum of risk rather than as a binary condition, with a significant number of moderate-income households facing affordability challenges. The policy choice matrix indicates that the highest-risk tier includes around 0.59 million households necessitating rapid intervention, whereas a more extensive moderate-risk group underscores the requirement for comprehensive stabilization initiatives.

The study illustrates that housing affordability stress in Canada may be accurately identified and prioritized through household-level data, facilitating a more efficient, equitable, and transparent distribution of housing assistance amid resource limitations.

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Chapter I: Introduction

1.1 Problem Identification

Housing affordability has become one of the most enduring socio-economic concerns in Canada in the last decade. Although escalating housing prices and rental expenses regularly prevail in public discussions, policy solutions are typically informed by aggregate metrics such as national average home prices, vacancy rates, or overall shelter-cost burdens that conceal significant variability among households. The outcome is a discrepancy between overarching housing narratives and the specific reality encountered by various household types.

In Canada, housing affordability stress is officially defined by the concept of Core Housing Need (CHN), as articulated by the Canada Mortgage and Housing Corporation (CMHC). This condition arises when a household's dwelling is deemed inadequate, unsuitable, or unaffordable, necessitating that the household allocate 30% or more of its before-tax income to secure acceptable alternative housing (CMHC, 2023). The affordability aspect is intricately linked to the shelter-cost-to-income ratio (STIR), with the 30% level functioning as a commonly accepted standard in Canadian housing studies. Despite criticisms regarding its rather arbitrary nature (Hulchanski, 2010; Stone et al., 2011), this criterion continues to be crucial in the formulation of federal and provincial housing programs.

Recent Canadian research repeatedly reveals significant variations in housing stress across different tenures and income levels. Renters are overrepresented among households experiencing significant housing demand, especially in major urban centers like Toronto and Vancouver (Zhu et al., 2023). Longitudinal evaluations indicate that affordability challenges have escalated more swiftly for renters compared to homeowners, highlighting constricting rental markets, financialization tendencies, and disparate income growth (August 2022; Zhu et al., 2023). Moreover, socioeconomic gradients in housing stress are consistent: lower-income households demonstrate significantly elevated STIRs and a greater prevalence of core housing needs in comparison to middle- and upper-income groups (Pomeroy & Lampert, 2019).

Regional disparities further complicate the national landscape. Major metropolitan regions have significant affordability challenges, whereas smaller urban centers and specific provinces have encountered steep rental inflation post-pandemic, without corresponding salary increases (CMHC, 2024). These regional inequalities indicate that housing affordability stress cannot be sufficiently comprehended or resolved through national averages alone.

Notwithstanding substantial descriptive evidence, a significant portion of the Canadian housing literature continues to be either macro-level or descriptive in nature. Numerous research depends on cross-tabulations, province aggregates, or metropolitan-level information (Leon & Iveniuk, 2021; Zhu et al., 2023). Although beneficial, these methods do not yield instruments that can specifically indicate which households should be prioritized in situations of resource scarcity. Housing agencies and policymakers encounter stringent budgetary constraints and capacity limitations. In this context, the challenge lies not only in recognizing significant structural discrepancies but also in devising a method for distributing aid that reduces unmet needs.

This study bridges the gap by transitioning from aggregate descriptions to household-level risk identification with the 2022 Canadian Housing Survey (CHS) Public Use Microdata File (PUMF). The approach employs interpretable logistic regression to model the possibility of a household experiencing core housing need (PCHN), offering a decision-support framework instead of a causal explanation. The objective is not to assert that income, tenure, or household composition "cause" housing stress, but to find observable indicators correlated with increased risk.

Household-level modeling is essential for three interconnected reasons.

Initially, policy targeting necessitates specificity. Programs like rent supplements, housing allowances, or support services are provided to households rather than provinces. Aggregated data cannot distinguish between two homes in the same area who have markedly different risk profiles due to factors such as tenure status, income level, or family makeup.

Secondly, risk identification facilitates triage in conditions of scarcity. When housing assistance is insufficient to cover all households in need, policymakers must establish operational criteria to assess eligibility. A calibrated probability model enables policymakers to modify thresholds based on budget limitations, explicitly balancing recall (identifying distressed households) with precision (reducing misallocation).

Third, interpretability is crucial in public-sector decision-making contexts. Black-box models may provide slight improvements in predicting accuracy but compromise transparency and accountability. Logistic regression offers coefficient-level interpretability, allowing stakeholders to comprehend the direction and relative significance of predictors. This is particularly significant in socio-economic policy situations, where equity, transparency, and public confidence are essential.

The Canadian Housing Survey 2022 PUMF provides an exceptionally appropriate dataset for this objective. The survey yields nationally representative household-level data regarding tenure, income, household composition, education, employment, and geography, as recorded in the project's data preparation phase. The modeling phase demonstrates that the outcome variable PCHN_Clean can be predicted with robust discriminatory performance (ROC-AUC = 0.90) while ensuring calibration appropriate for policy interpretation. Survey weights (PFWEIGHT) are designated for descriptive analysis rather than model training, ensuring representativeness in reporting and preventing distortions in the estimate process.

Calibration and threshold selection are not mere technical considerations but rather significant policy-related judgments. The modeling log indicates that lowering the classification threshold from 0.50 to 0.30 significantly enhances the recall of stressed families, hence broadening coverage but incurring more false positives. This demonstrates that risk modeling in housing policy is fundamentally normative: the choice of an operational threshold dictates who is identified for assistance. By meticulously recording these trade-offs, the project harmonizes statistical modeling with practical software limitations.

In summary, the primary issue examined in this capstone is not merely the existence of housing affordability stress in Canada this is widely established but rather that current aggregate methodologies offer insufficient operational direction for prioritizing households amid resource limitations. This study enhances a decision-support framework based on Canadian housing research and microdata by integrating weighted descriptive analysis with interpretable logistic regression and calibrated threshold selection.

1.2 Research Question and Hypothesis

This project aims to address the stated disparity between descriptive housing research and operational targeting requirements by answering the following primary question:

What household-level factors are most significantly linked to core housing demand in Canada, and how might these relationships be implemented into a transparent, calibrated decision-support system given constrained housing resources?

The study centers on PCHN as the principal binary outcome, with structural predictors comprising tenure, income, family composition, presence of children, education, occupation, and province.

The descriptive hypotheses are based on recognized Canadian literature:

1. Renters are considerably more prone than homeowners to have fundamental housing need.

2. Lower-income households demonstrate a greater likelihood of experiencing core housing need than middle- and high-income households.
3. The incidence of core housing requirement significantly differs among provinces and areas.

The predictive hypothesis transcends mere description:

1. A concise and interpretable logistic regression model utilizing observable household variables can effectively identify homes at increased risk of core housing need, facilitating calibrated decision thresholds appropriate for policy targeting.

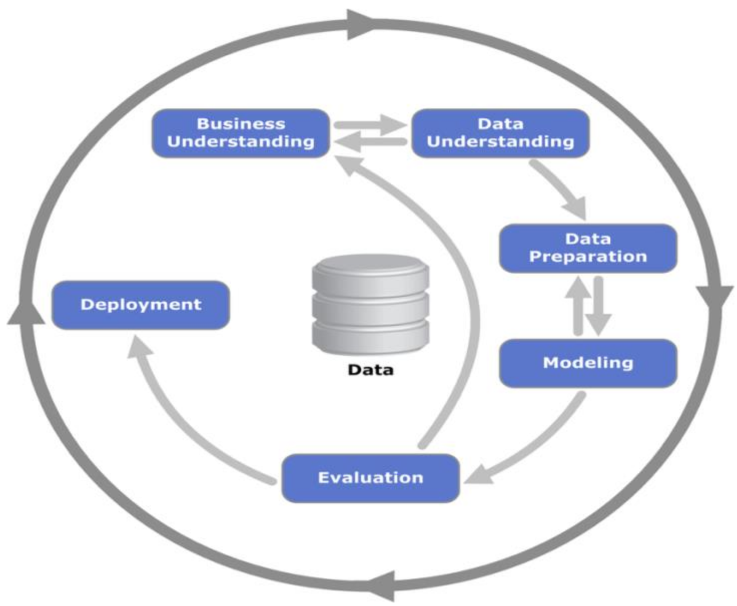
The hypotheses are assessed via weighted descriptive statistics and a stratified train-test logistic regression model. The focus is mostly on prediction validity, interpretability, and calibration, rather than on causal inference. The resultant approach is intended to facilitate evidence-based allocation decisions while ensuring transparency and defensibility in public policy settings.

Chapter II: Data Collection and Preparation

2.1 Data Collection

This study adheres to a structured analytics lifecycle aligned with CRISP-DM principles (Shearer, 2000), advancing from data comprehension and preparation to predictive modeling, validation, and decision-support implementation.

Figure 1Overview of the CHS 2022 Data Analytics Lifecycle: From Data Preparation to Decision Support



The principal dataset utilized in this research is the Statistics Canada Canadian Housing Survey (CHS) 2022 Public Use Microdata File (PUMF). The CHS is a nationally representative cross-sectional survey aimed at gathering comprehensive data on housing conditions, affordability, tenure, household demographics, and other socio-economic factors throughout Canada.

The 2022 PUMF has 38,657 household-level data and 103 variables, as recorded during the exploratory phase. The analytical unit is the private household. The dataset contains survey weights (PFWEIGHT) to guarantee national representativeness in the generation of descriptive statistics.

The variable of interest is the outcome: **Primary Household in Core Housing Need (PCHN)** Core Housing Need is delineated by the criteria of adequacy, suitability, and affordability according to CMHC rules. In this study, PCHN functions as a binary outcome indicating housing affordability stress at the household level.

Key structural predictors extracted for analysis include:

1. Tenure (PDCT_05_Clean: Owner vs Renter)
2. Total household income (PHHTTINC)
3. Derived income quintile
4. Household size (PHHSIZE)
5. Household type (PHTYPE)
6. Presence of minors (PMINOR)
7. Education level (PHGEDUC)
8. Employment presence (PEMLP)
9. Province and regional indicators
10. Survey weight (PFWEIGHT)

The CHS PUMF is especially appropriate for this research as it facilitates micro-level analysis while preserving nationwide coverage. In contrast to aggregated census summaries, the microdata structure facilitates household-level modeling essential for risk identification and decision-making support.

The PUMF is recognized as anonymized, with geographic identifiers consolidated into larger regions, hence restricting targeting at the municipal level. Nonetheless, the existing framework for provincial and regional policy analysis remains suitable and statistically sound.

2.2 Data Cleaning

The data cleaning process adhered to a systematic and replicable approach recorded throughout the EDA phase.

Survey datasets often include reserved or placeholder codes denoting:

1. Not specified (e.g., 9)
2. Not relevant (e.g., 99)
3. Excessive placeholder values (e.g., 99999999999 for income)

Instead of eliminating these observations entirely, reserved codes were transformed into missing values (NA). This method maintains the integrity of the denominator and prevents false inflation or distortion of prevalence rates.

Variables were generated with a _Clean suffix to ensure complete repeatability and distinction from raw fields.

2.3 Construction of Analytical Datasets

An analytical framework (df_analytic) was established after the cleaning methods. All descriptive analysis and hypothesis testing were conducted using this organized dataset to maintain denominator consistency between comparisons.

The modeling dataset was divided into training and testing subsets by stratified sampling to maintain the proportions of outcome classes.

This systematic framework guarantees openness and repeatability while mitigating the dangers of data leakage.

2.4 Application of Survey Weights

Survey weights (PFWEIGHT) were applied solely in descriptive analysis to generate nationally representative estimates of housing stress prevalence.

Weighted estimates indicate that approximately 11.39% of Canadian households experience core housing need**, r**eflecting the nationally representative prevalence after applying survey weights. The unweighted sample proportion in the CHS microdata is approximately 16.53%**,** indicating that the raw sample slightly overrepresents households experiencing housing stress.

Weights were deliberately excluded during the calculation of logistic regression. This selection signifies a focus on predictive modeling rather than estimating population parameters. Employing weights in supervised learning can affect probability calibration and undermine discrimination performance when the primary objective is risk identification rather than inference.

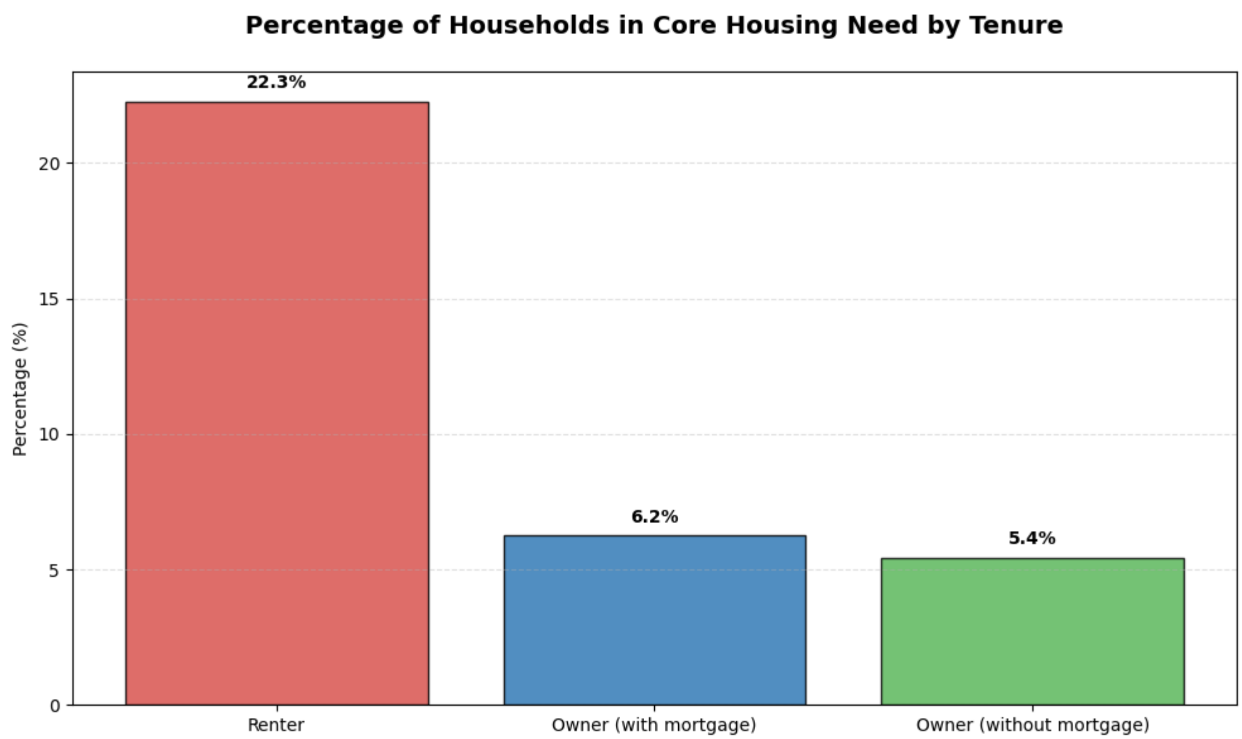
This contrast embodies a crucial principle in applied analytics: representativeness is vital for descriptive estimates, while predictive modeling emphasizes discrimination and calibration within the observable data structure.

Chapter III: Statistical Analysis and Diagnostics

3.1 Descriptive Analysis and Hypothesis Testing

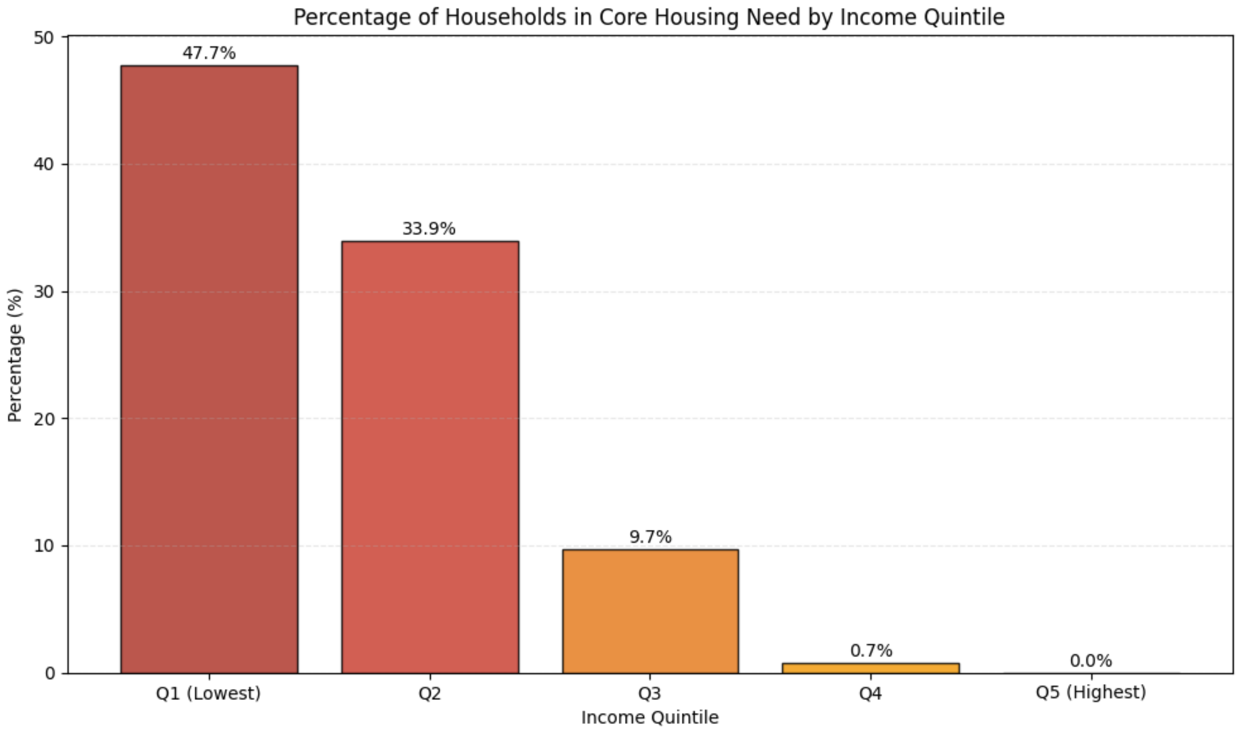
Before proceeding to predictive modeling, a descriptive study was performed to establish the empirical framework of housing affordability stress within the CHS 2022 data and to evaluate the initial three hypotheses utilizing survey-weighted statistics. This step was essential as the prediction phase aimed to enhance observable household-level patterns rather than supplant them. This descriptive phase, along with the project's analytical purpose, concentrated on association, distribution, and group differences, refraining from asserting causal relationships. Survey weights (PFWEIGHT) were utilized during the descriptive analysis to guarantee that estimates accurately represented the national household distribution in Canada. The weighted baseline data indicated that approximately 11.39% of Canadian households experience core housing need**,** providing a nationally representative estimate after applying survey weights. This discovery confirmed the national significance of the issue and the mild class imbalance that would subsequently impact predictive assessment. The first descriptive hypothesis (H1) posited that renter households are more likely than owner households to experience housing affordability stress. This hypothesis was examined using survey-weighted cross-tabulations between tenure status and multiple indicators of housing stress, including core housing need (PCHN_Clean), reported financial difficulty related to housing costs, and indicators of acute payment stress. The results consistently showed that renter households exhibited substantially higher levels of housing stress compared to owner households. Specifically, renters were more frequently classified as being in core housing need, reported greater difficulty managing housing costs, and were more likely to indicate skipped or delayed housing payments. Taken together, these descriptive findings provide strong evidence in support of Hypothesis 1 and justify the inclusion of tenure status as a key structural predictor in the subsequent predictive modeling stage.

Figure 2 Weighted Housing Affordability Stress by Tenure (Owner vs. Renter)



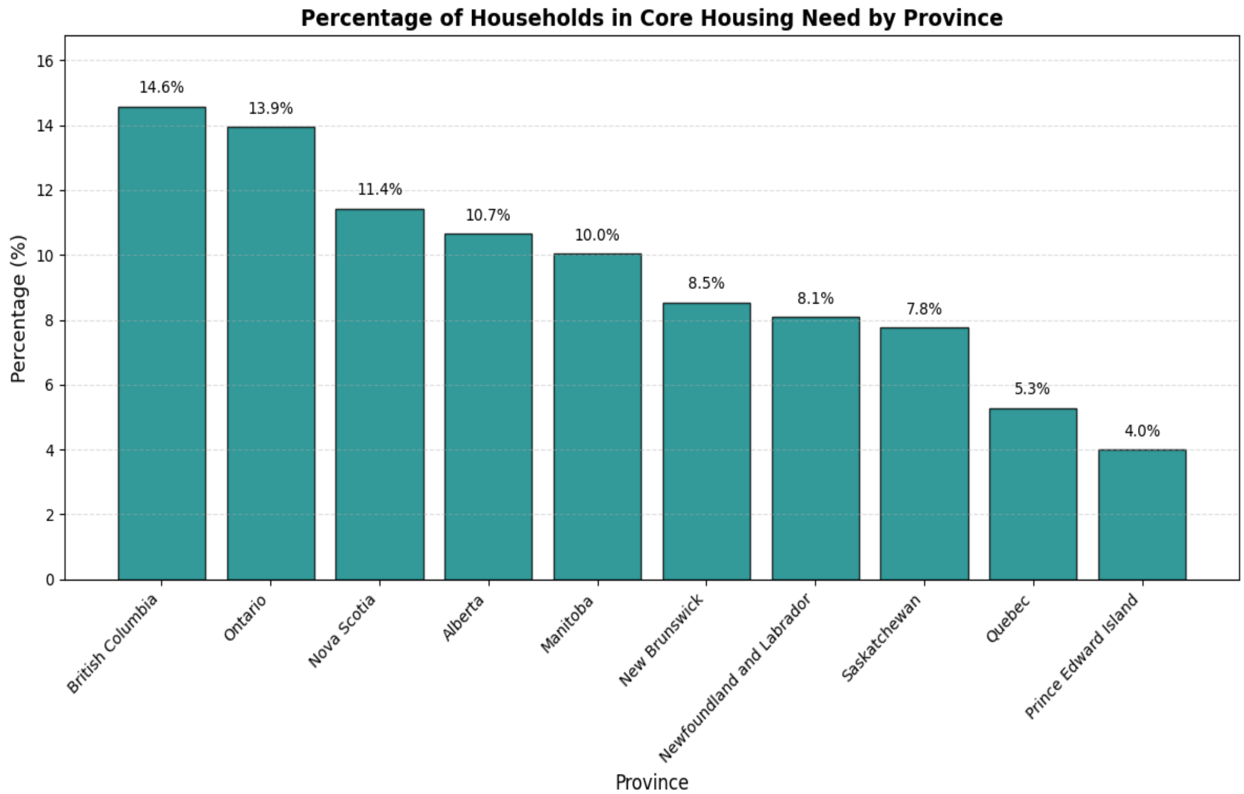
The second descriptive hypothesis (H2) proposed that lower-income households endure greater home affordability stress compared to middle- and higher-income households. Households were categorized into income quintiles and analyzed in relation to core housing needs and associated financial stress indicators by survey-weighted cross-tabulations. A distinct monotonic gradient was noted: households in lower-income quintiles faced significantly elevated housing stress, whereas stress diminished steadily with increasing income. This gradient was seen not only for fundamental housing requirements but also for self-reported challenges with affordability and instances of missed payments. The consistency of this trend robustly corroborated Hypothesis 2 and validated income as a critical variable of affordability risk within the Canadian context.

Figure 3 Weighted Housing Affordability Stress by Income Quintile



The third descriptive hypothesis (H3) posited that housing affordability stress differs among provinces and regions. This was evaluated by weighted comparisons of PCHN_Clean across geographic groups. The results demonstrated significant variance in the proportion of households experiencing core housing need among provinces and regions, even after eliminating reserved codes and employing a uniform analytical framework. These disparities indicate that home affordability stress is influenced by both household economic status and wider spatial and market factors. Consequently, Hypothesis 3 was validated, and geography was maintained as a significant contextual variable for further research.

Figure 4 Weighted Core Housing Need by Province/Region



The descriptive findings collectively provide empirical validation for the initial three hypotheses. Moreover, they formulated the structural rationale for the subsequent predictive phase. Tenure, income, and geography were not chosen arbitrarily as model inputs; instead, they were maintained according to the weighted descriptive analysis indicating a significant correlation with housing affordability stress in the Canadian household demographic. Simultaneously, these descriptive findings are confined to observed correlations. Their analysis does not clarify if the same indicators can consistently identify household-level risk when evaluated collectively within a predictive model. The subsequent section addresses the question with interpretable logistic regression.

3.2 Implications of Descriptive Findings for Model Design

Section 3.1's descriptive analysis identified various structural patterns in the distribution of housing affordability stress among Canadian households. These findings play a crucial function beyond mere description; they guide the design and specification of the predictive modeling phase. The modeling framework was developed in conjunction with empirical trends identified during the exploratory analysis. The pronounced tenure gradient identified in the descriptive analysis warranted the use of housing tenure (PDCT_05_Clean) as a principal predictor in the modeling phase. The higher level of core housing demand among renters indicates that tenure status reflects exposure to varying housing market dynamics, especially within the rental sector, where households may be more susceptible to cost variations. The descriptive data imply a connection, not causation, suggesting that tenure status is a significant indicator for identifying households at increased risk of housing affordability stress. The consistent gradient seen across income quintiles reinforces the necessity of include household income (PHHTTINC_Clean) as a fundamental structural predictor. The data indicate that housing stress diminishes progressively with increasing income, demonstrating that economic capacity is a fundamental aspect of housing affordability. In the prediction framework, income serves as a crucial variable for differentiating households with varying degree of financial assistance to housing expenses. The observed difference in housing stress between provinces and regions justifies the incorporation of geographic indicators (Province_Name) into the modeling framework. The housing markets in Canada exhibit considerable variation in cost structures, supply limitations, and economic situations. Incorporating geography enables the prediction model to account for contextual disparities without supposing the fundamental causes influencing regional variance. In addition to variable selection, the descriptive analysis indicated that the outcome variable core housing need demonstrates a moderate class imbalance, with approximately 11.39% of households experiencing housing stress in the weighted population estimates . This disparity has significant ramifications for predictive modeling. In classification tasks with imbalanced outcome distributions, conventional accuracy metrics may yield deceptive evaluations of model efficacy. Consequently, evaluation metrics including ROC-AUC, accuracy, and recall serve as more useful indicators of a model's proficiency in accurately identifying the minority class. The descriptive analysis establishes a systematic basis for the predictive modeling phase. The chosen variables for modeling tenure, income, household composition, education, employment, and geography were retained due to their availability in the dataset and the exploratory analysis indicating their significant correlation with housing affordability stress in Canada. This stage strengthens methodological rigor by ensuring that predictive modeling is based on experimentally established associations rather than solely on exploratory feature selection. The subsequent section transitions from descriptive association to predictive analysis. This assessment specifically determines if the identified household factors can collectively forecast the likelihood of a household experiencing core housing need, hence addressing the project's fourth hypothesis about the viability of risk identification using interpretable predictive modeling.

3.3 Transition to Predictive Modeling

The descriptive research established that tenure, income, and geographic location are significantly correlated with home affordability stress among Canadian households. Although these findings delineate significant structural links, they do not ascertain whether these features may be employed to identify home at risks on an individual basis. This analysis advances to a predictive modeling phase aimed at assessing the fourth hypothesis (H4) of the study: that discernible household features can be utilized to identify homes at increased risk of core housing need. The objective of this phase is to identify risks for decision-making support, rather than to establish causal relationships.

A logistic regression model that is interpretable was chosen as the baseline prediction method due to its ability to generate probabilistic risk estimates that can be clearly understood and utilized in subsequent decision-support tools.

Approximately 11.39% of households experience core housing need in the weighted population estimates**,** while the raw sample prevalence in the survey data is approximately 16.53%, resulting in a moderate class imbalance in the outcome variable. Consequently, model evaluation emphasizes measures for discrimination and minority-class identification rather than solely accuracy.

This section delineates the modeling framework and evaluation methodologies employed to determine prediction efficacy.

3.4 Methods Applied

To evaluate whether household variables can forecast the probability of housing affordability stress, two supervised classification methods were analyzed: Logistic Regression and Random Forest. Logistic regression was deemed the principal modeling method because of its interpretability, stability in binary classification tasks, and appropriateness for policy-oriented decision-support contexts. A Random Forest model was also employed as a benchmark to evaluate if a non-linear ensemble strategy may yield superior prediction performance. While ensemble models can clarify complex relationships, logistic regression offers clear coefficient estimates that enable policymakers to comprehend the impact of individual household factors on housing stress risk. This interpretability is especially significant in policy scenarios where model transparency and elucidation are crucial.

The modeling dataset was derived from the sanitized analytical framework (df_analytic) created during the data preparation phase. The dependent variable used for prediction was PCHN_Clean, designated as 1 for households experiencing core housing need (housing stress) and 0 for those not in need, in accordance with the CHS codebook.

Feature Set: Some of the ultimate modeling feature set comprised essential household-level structural variables uncovered in the descriptive analysis phase:

1. PDCT_05 – housing tenure (ownership versus rental)
2. PHHTINC – aggregate household income
3. PHTYPE – domestic structure
4. PMINOR – presence of minors in the residence
5. PHGEDUC - educational attainment of the household reference individual
6. PEMPL – employment presence in the home
7. Province_Name — geographical control variable

The selected variables were identified due to their significant connections with housing affordability stress, as evidenced by the exploratory analysis, while also ensuring interpretability and relevance to policy.

Train-Test Split

The dataset was divided using a stratified 80/20 train-test split, maintaining a consistent proportion of households in core housing need throughout both training and testing subsets.

The resulting partition comprised:

1. Training observations: 29,352 households
2. Testing observations: 7,076 homes

The positive class (households in core housing need) included around 16.5% of the observations in both the training and test datasets. Stratification guaranteed that the model acquired knowledge from a representative distribution of the outcome variable

Data Preprocessing Pipeline

A preprocessing pipeline was established to guarantee uniform handling of numerical and categorical variables during model training. The pipeline comprised:

1. Categorical variables were imputed with a "Missing" category and encoded by one-hot encoding, excluding the first category to prevent multicollinearity.
2. Numerical variables: imputed with median values and standardized using feature scaling

This organized pipeline eliminates information leakage between training and testing datasets and guarantees that preparatory treatments are uniformly executed.

Model Specification

Logistic Regression and Random Forest classifiers were used for predictive assessment. Logistic regression functioned as the principal interpretable model, whilst Random Forest was employed as a benchmark to evaluate predictive performance inside a non-linear ensemble framework.

Logistic regression was configured with the following settings:

1. Solver: saga
2. Penalty: L2 Regularization
3. Maximum iterations: 5,000

This configuration was chosen for its reliable performance with high-dimensional one-hot encoded categorical variables, while preserving interpretable coefficient estimates.

Notably, survey weights (PFWEIGHT) were excluded from the model training process. The weights were allocated for descriptive analysis and interpretation. In predictive modeling, the aim is to identify risks rather than estimate population parameters, and the application of weights during training can skew probability calibration.

The trained model generates a predicted probability for each household, indicating the expected possibility of the home experiencing core housing need. These probabilities underpin future model review, threshold analysis, and downstream decision-support applications.

3.5 Model Performance

After model training, the accuracy of predictions was assessed using measures appropriate for binary classification characterized by significant class imbalance. Since only about 16.5% of households in the dataset have fundamental housing need, traditional accuracy measurements would yield a deceptive evaluation of model efficacy. Both Logistic Regression and Random Forest models were assessed utilizing discrimination criteria such as Receiver Operating Characteristic Area Under the Curve (ROC-AUC), Precision-Recall Area Under the Curve (PR-AUC), recall, and precision.

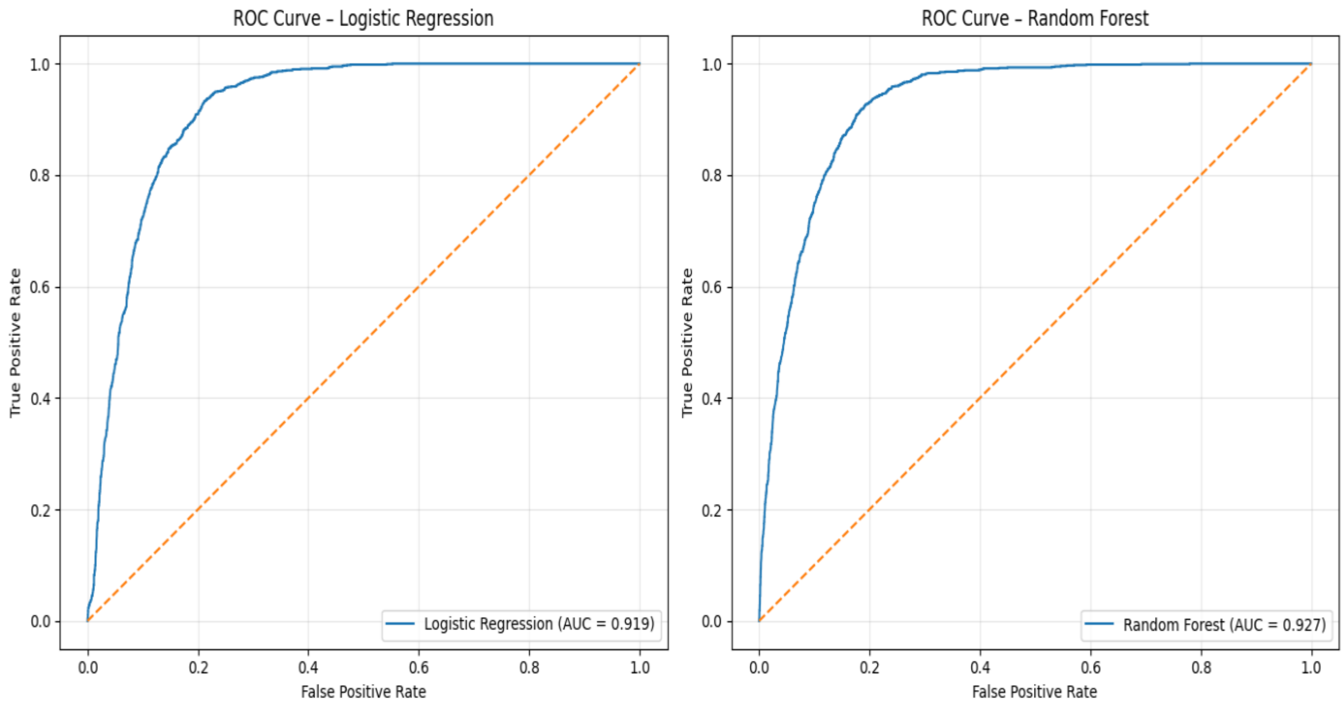
The Random Forest model demonstrated marginally superior predictive discrimination, with a ROC-AUC of roughly 0.927, in contrast to 0.919 for logistic regression. The increase in prediction performance was rather slight.

Discriminatory Performance

The key metric for model discrimination was the Receiver Operating Characteristic Area Under the Curve (ROC-AUC). This score assesses the model's capacity to differentiate between households in core housing need and those not in need across all potential classification levels.

The logistic regression model attained a ROC-AUC of 0.919, signifying robust differentiation between stressed and non-stressed households. An ROC-AUC over 0.90 indicates that the model effectively extracts significant predictive information from the household variables within the feature set.

Figure 5 ROC Curve for Model Predicting Core Housing Need



The Precision–Recall Area Under the Curve (PR-AUC) was further evaluated to evaluate minority-class detection performance. The model achieved a PR-AUC of 0.5991, significantly above the baseline prevalence of the positive class, hence demonstrating a notable improvement in the identification of stressed households compared to random classification.

Default Threshold Performance

Using the conventional classification threshold of 0.50, the model produced the confusion matrix shown below.

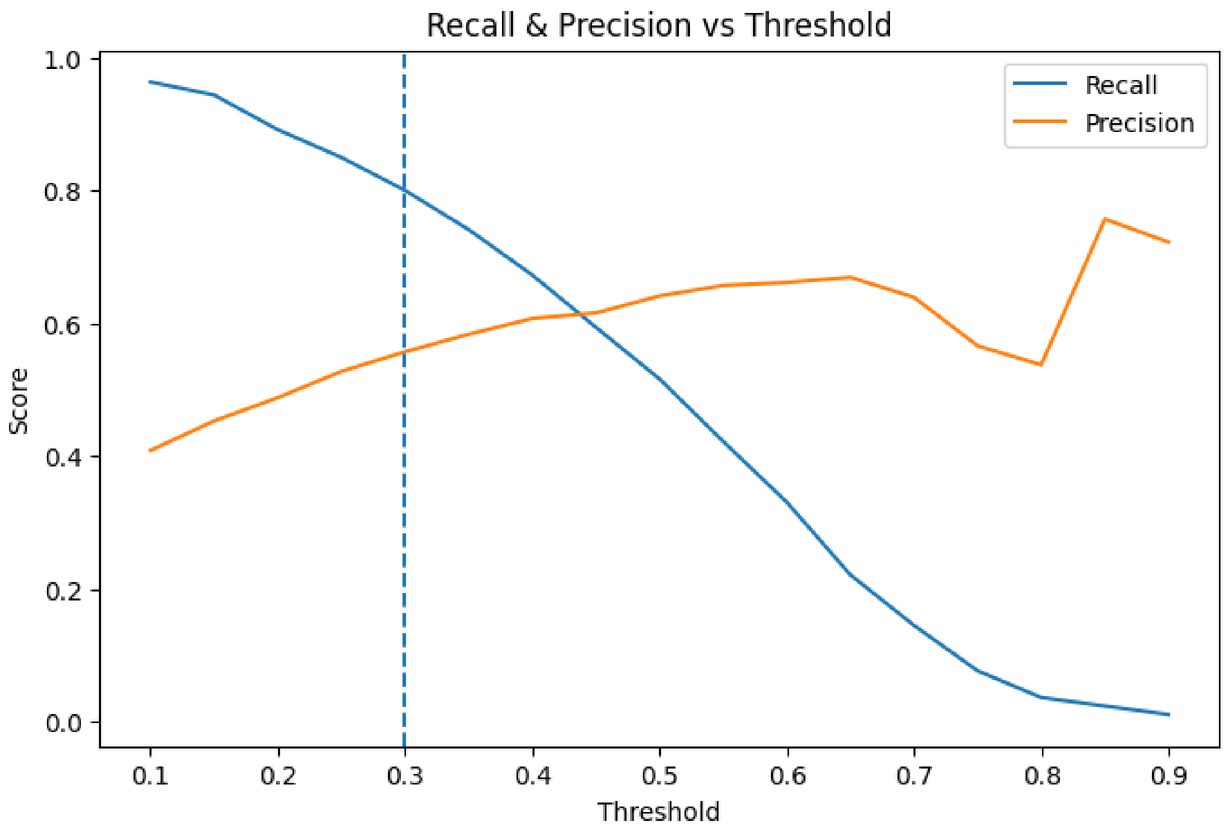
Table 1 Confusion Matrix at Threshold = 0.50

	Predicted: Not Stressed	Predicted: Stressed
Actual: Not Stressed	5568	338
Actual Stressed	566	604

At this threshold, the model accurately detects several non-stressed households but captures just a little above fifty percent of stressed households, yielding a recall of roughly 0.5162 and a precision of approximately 0.6412. This behavior illustrates the conservative characteristic of the default threshold in imbalanced classification issues, where models typically prioritize the majority class. Threshold Optimization for screening. The objective of this study is to identify risks for decision support; hence, threshold selection was assessed as an operational design choice rather than a static statistical parameter. A threshold sweep was conducted over various probability cutoffs to analyze the trade-off among recall, precision, and the proportion of households identified for potential intervention.

When applying the selected 0.30 operational threshold, the model flags approximately 10.52% of households in the weighted population distribution as being at elevated risk of housing stress. This figure is slightly below the estimated weighted prevalence of 11.39%, suggesting that the screening configuration captures a substantial portion of at-risk households while maintaining manageable administrative caseloads.

Figure 6 Precision and Recall Across Probability Thresholds



Operating Performance at Threshold 0.30

Using the selected screening threshold of 0.30, the confusion matrix becomes:

Table 2 Confusion Matrix at Threshold = 0.30

	Predicted: Not Stressed	Predicted: Stressed
Actual: Not Stressed	5162	744
Actual Stressed	234	936

This operating point yields the subsequent performance characteristics:

1. Recall (Stress Detection Rate): about 0.8
2. Precision: about 0.557
3. Flagged Household Rate: approximately 23.7% of the test dataset

The results demonstrate that the model accurately identifies around seventy-five percent of households undergoing housing stress, while restricting the number of households designated for further assessment to a manageable proportion of the population.

Interpretation for Decision Support

This threshold serves as a screening or triage mechanism rather than a conclusive classification criterion from a policy standpoint. The objective is to identify a cohort of homes with higher probable risk for subsequent evaluation using decision-support methods, like segmentation analysis, rule extraction, or resource allocation simulations.

Applying probabilistic risk scores enables policymakers to modify thresholds based on available resources or program capacity. A reduced threshold may be employed in initiatives aimed at encompassing a maximum number of disadvantaged households, whereas a higher threshold may be suitable when administrative resources are constrained.

This part evaluates diagnostic validation and robustness tests to enhance the trustworthiness of the model's probability estimations prior to their implementation in subsequent analytical processes.

3.6 Diagnostic Validation and Robustness Checks

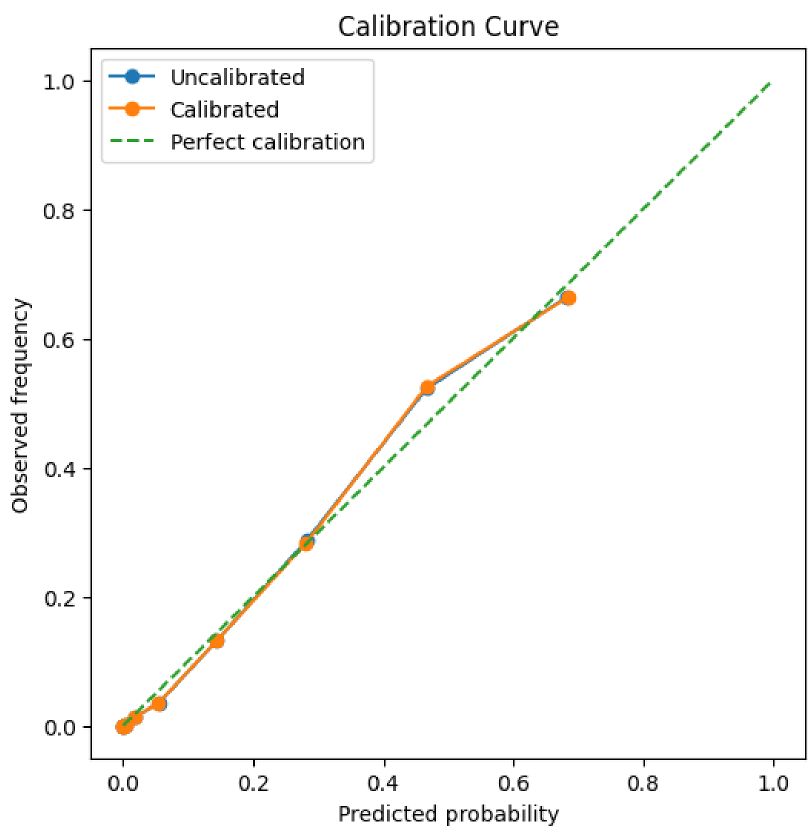
Beyond discrimination performance, it is crucial to determine the reliability of the model's probability estimations and their alignment with external measures of housing affordability stress. Diagnostic validation was accomplished using probability calibration analysis and a robustness comparison with an independent affordability metric.

Probability Calibration

Calibration assesses the alignment between anticipated probability and actual outcome frequencies. If a cohort of households is assigned a predicted probability of 0.30, around 30% of those families ought to be experiencing core housing need for the model to be deemed effectively calibrated.

Calibration was assessed by categorizing anticipated probabilities into 10 quantile-based bins, and the mean predicted probability within each bin was juxtaposed with the actual rate of housing stress.

Figure 7 Calibration Curve for Predicted Housing Stress Probabilities



The results demonstrate that predicted probability align closely with observed outcome rates within the decision-relevant range of the distribution. Bins associated with moderate and higher expected risk levels exhibit a strong correlation between predicted and observed stress rates. This indicates that the model's probability estimations are adequately credible for application in subsequent decision-support tools.

Calibration is crucial in policy-oriented predictive modeling as risk scores are frequently employed to inform prioritizing decisions instead of generating binary classifications. Accurately calibrated probabilities enable policymakers to comprehend expected risk levels in significant terms and modify intervention thresholds based on available program capacity.

External Robustness Check Using Shelter Cost Burden

As an additional validation step, model predictions were compared with PSTIR_GR, an independent affordability indicator denoting shelter-cost-to-income ratio classifications. This variable was deliberately excluded as a predictor in the model, so guaranteeing that the comparison serves as an external consistency check rather than a circular validation process, Households were grouped by PSTIR category, and the following statistics were calculated for each group:

1. number of households
2. observed core housing need rate
3. average predicted risk score
4. proportion of households flagged at the selected threshold

Table 3 Robustness check

	PSTIR_GR	n_households	true_stress_rate	predicted_risk_mean	flagged_rate
0	1.0	5443	0.026272	0.104675	0.135036
1	2.0	1265	0.581818	0.337660	0.532806
2	3.0	368	0.790761	0.466136	0.736413

Results indicate a consistent link between shelter-cost burden and anticipated housing stress. Households with elevated shelter-cost-to-income ratios demonstrate increased rates of core housing need and higher anticipated probabilities according to the model. This alignment suggests that the predictive method is effectively capturing significant affordability aspects without explicitly depending on the shelter-cost ratio as an input variable.

Summary of Diagnostic Findings

The diagnostic checks, when looked at together, show that the predictive model is reliable. The model shows that it can tell the difference between things very well, that it can calibrate probabilities well, and that it always agrees with an independent affordability indicator. These findings indicate that the predicted probabilities offer a reliable foundation for subsequent analytical applications.

Once predictive validation is done, the analysis moves on to the analytics application phase. This is where the model outputs are turned into decision-support tools that can be understood, like risk-based segmentation and rule extraction.

Chapter IV: Analytics Application

4.1 Overview of the Analytics Application Stage

Although the predictive modeling phase generated reliable assessments of housing affordability risk at the household level, likelihood scores by themselves are not immediately actionable for policymakers. Housing authorities have to translate analytical results into operational judgments concerning the allocation of support, the distribution of limited resources, and the suitability of policy initiatives for various household groups.

To address this weakness, the project implements a variety of analytical applications that convert prediction results into actionable decision-support instruments. These applications function across multiple interrelated analytics subfields. Initially, general analytical techniques are employed to enhance interpretability by converting continuous model predictions into comprehensible household profiles. Secondly, operational and prescriptive analytics methodologies are employed to model policy decisions within the confines of real-world limitations, including constrained program budgets and administrative capabilities.

The analytics application phase thus functions as the conduit between predictive modeling and policy execution. This stage implements the findings of the logistic regression model by categorizing households based on risk profiles, deriving clear targeting criteria, and testing various policy allocation techniques.

Two supplementary analytical methods are initially presented to facilitate the interpretation of the model outputs. Risk-based segmentation categorizes households with analogous features into comprehensible risk profiles, whereas decision rule extraction converts model predictions into clear if–then targeted directives. These technologies enhance the interpretability of the predictive system and assist policymakers in comprehending the structural patterns that underlie housing affordability stress.

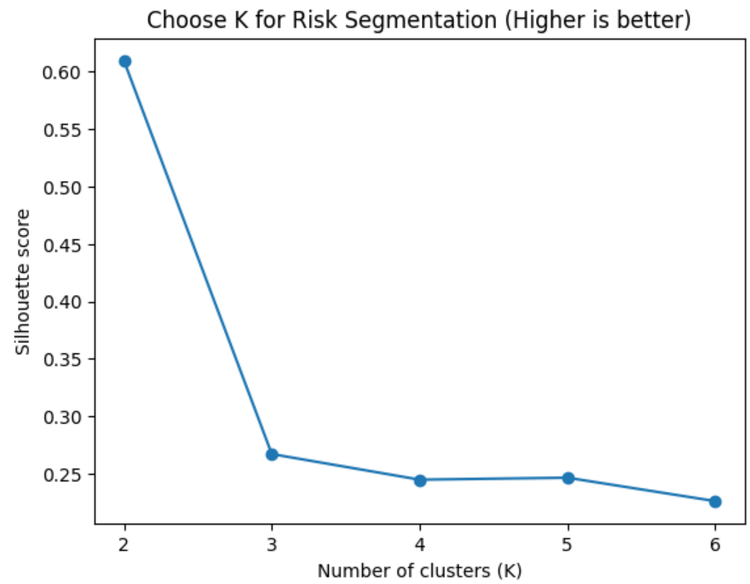
The primary contribution of this phase is the prescriptive analytics framework, which assesses the optimal allocation of housing support resources within budgetary and capacity limitations. Utilizing the anticipated risk ratings from Chapter III, several policy simulations are performed to analyze the optimization of resource allocation, the trade-offs in administrative workload, and the comparative effectiveness of machine learning-based targeting vs conventional policy methods. Collectively, these analytical applications illustrate how predictive modeling transcends descriptive analysis to facilitate evidence-based housing policy decisions.

4.2 Risk-Based Segmentation

The predictive model offers probability estimates for each home; however, individual probability scores may pose interpretative challenges for policymakers when considered at scale. A risk-based segmentation study was performed to categorize homes with analogous features and anticipated risk levels for enhanced interpretability.

Segmentation was executed with K-Means clustering on the scored dataset (df_scored). Clustering is an unsupervised learning method that discerns inherent groupings in data by evaluating the similarity among observations. This study employed clustering based on predicted risk scores alongside key structural variables such as household factors, including estimated housing stress probability, tenure status, household income, family type, presence of children, and geographic location. The ideal number of clusters was ascertained by the silhouette score, which assesses the degree to which families conform to their designated clusters in comparison to alternative clusters. The study revealed three major clusters: high-risk, moderate-risk, and low-risk household characteristics.

Figure 8 4.1 Overview of the Analytics Application Stage

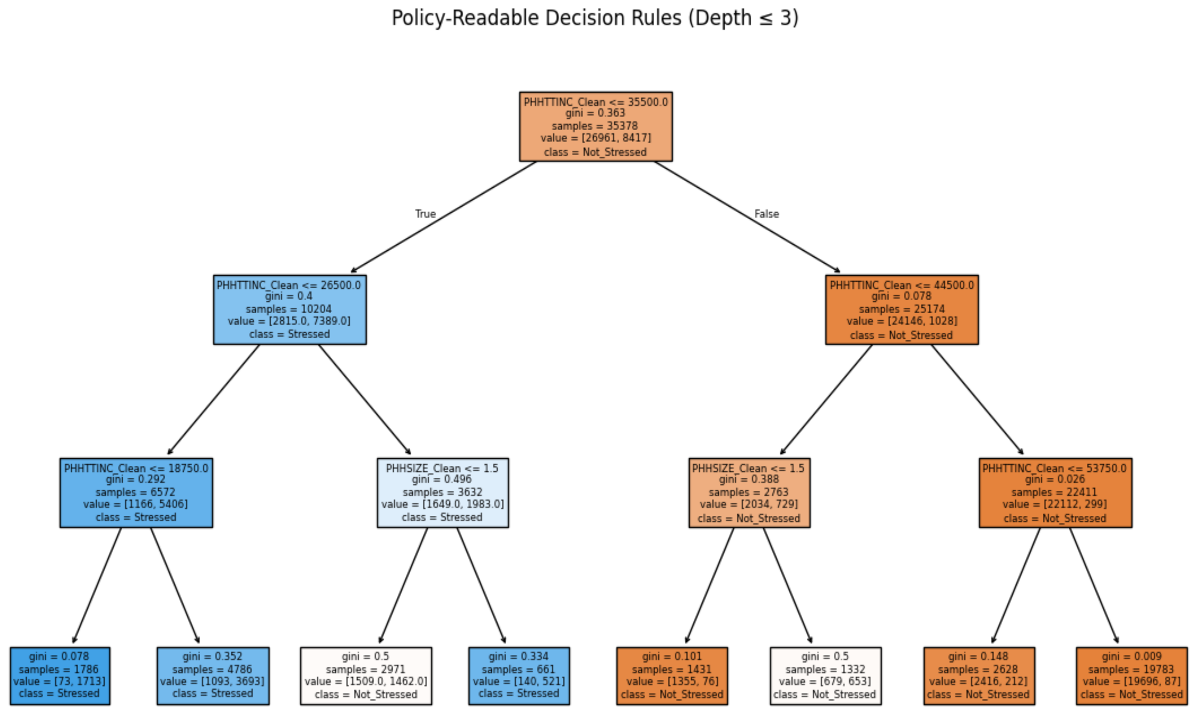


The resulting clusters demonstrated distinct structural disparities in housing risk. The high-risk cluster comprised a significant proportion of rental households with lower incomes and elevated estimated risks of housing stress. The moderate-risk cluster comprised mixed tenure households with intermediate income levels and a moderate anticipated risk. The low-risk cluster was primarily comprised of affluent homeowner households exhibiting minimal estimated housing stress risks.

4.3 Decision Rule Extraction

While clustering offers general household classifications, policymakers frequently want explicit targeting criteria that can be readily implemented in administrative contexts. A shallow decision tree was employed to derive transparent rules from the results of the predictive model.

Figure 9 Interpretable Decision Tree for Housing Stress Identification



The resulting tree indicates that household income is the principal division, with tenure (renter versus owner) emerging in the subsequent branches. This signifies that tenants within specific income brackets encounter significantly elevated anticipated housing stress risk.

These regulations establish a clear and comprehensible screening framework, converting the predictive model into criteria that policymakers can readily grasp when selecting households for housing assistance programs.

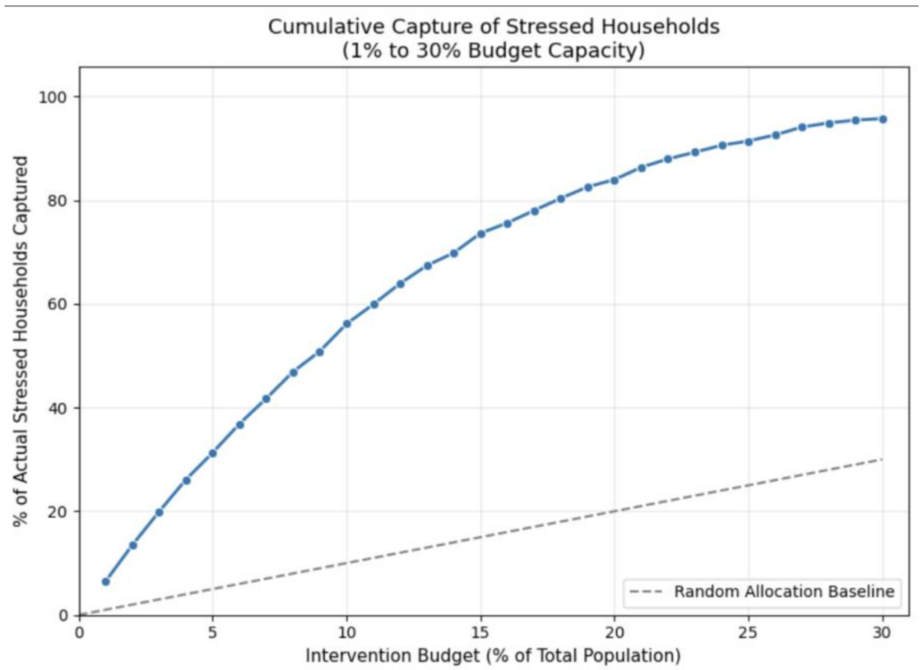
This section presents the prescriptive analytics framework, which assesses the utilization of anticipated risk scores for the more effective allocation of housing support resources.

4.4 Resource Allocation Optimization

The initial prescriptive analysis assesses the optimal allocation of housing aid under constrained program finances. Government housing initiatives seldom possess the capacity to assist all households facing affordability challenges, necessitating prioritizing. This strategy evaluates how households might be prioritized based on anticipated risk to optimize the fulfillment of housing demand within established coverage limits.

Households in the evaluated dataset were rated based on their projected chance of experiencing housing stress (prob_housing_stress). A weighted simulation utilizing the survey weight (PFWEIGHT) was subsequently performed to assess the extent of national housing demand that may be addressed by assisting the highest-risk households under various budget scenarios, spanning from 1% to 30% of the population.

Figure 10 Capture Rate of Housing Stress Under Risk-Based Allocation



The findings indicate that a risk-based allocation technique addresses housing needs significantly more effectively than random selection. The revised simulation indicates that directing resources to the top 30% of highest-risk homes encompasses roughly 95.7% of stressed households, illustrating the model's significant prioritization capability.

Initial program expansions yield the most significant enhancements in capture efficiency. For instance, augmenting coverage from 5% to 10% elevates the proportion of stressed households identified from 31.3% to 56.1%, signifying a marginal increase of around 5% more stressed families for each 1% increment in funding during the initial phases of intervention. As coverage increases, marginal profits progressively reduce, reaching 0.31% at approximately 30% coverage, exemplifying the anticipated trend of declining returns.

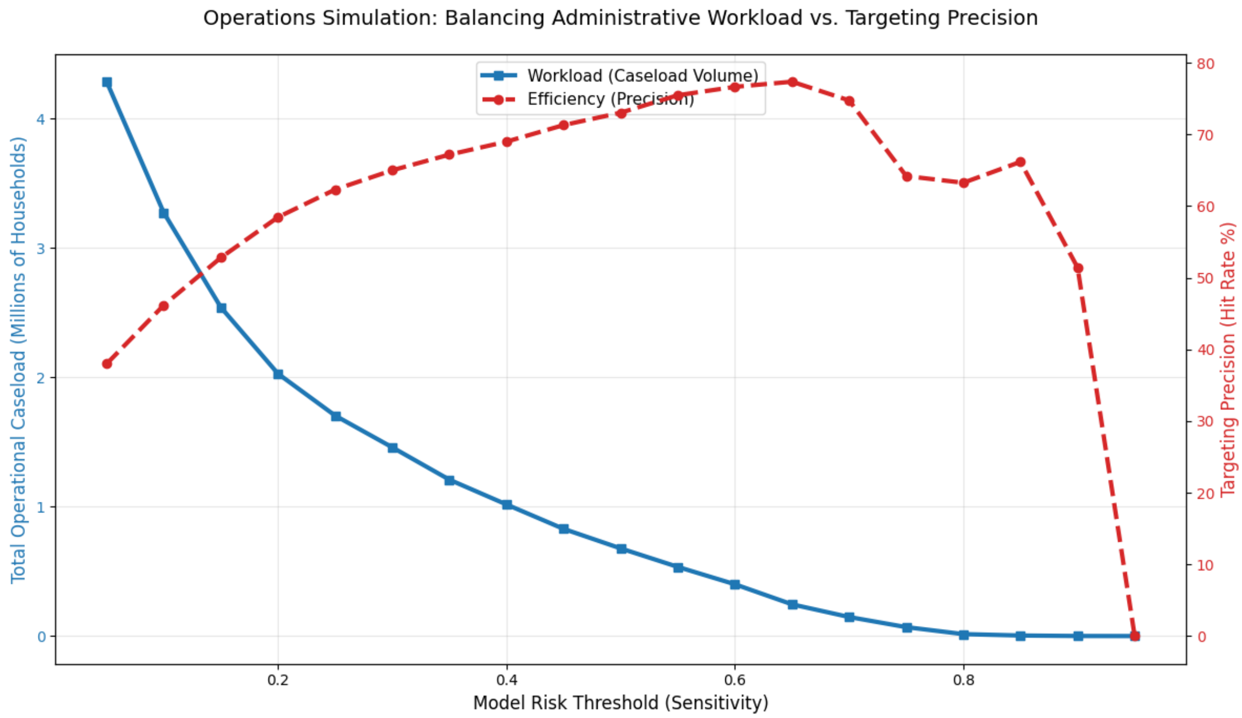
The findings indicate that risk-based prioritization significantly enhances the efficiency of housing support programs. Instead of allocating resources arbitrarily or on a first-come basis, policymakers might utilize anticipated risk scores to guarantee that constrained budgets address the maximum proportion of households facing housing affordability challenges.

4.5 Administrative Workload and Threshold Optimization

Although risk-based prioritizing enhances targeted efficiency, housing organizations must also account for limitations in administrative capability. A predictive model is operationally effective only if the quantity of households identified for intervention is manageable for program administrators.

A threshold simulation was performed to assess this trade-off utilizing the projected probability scores (prob_housing_stress). Various probability thresholds were evaluated to ascertain the number of homes that would be identified for review in each scenario. Survey weights (PFWEIGHT) were employed to transform these data into estimates of the total number of Canadian households for administrative review.

Figure 11 Administrative Caseload Under Alternative Risk Thresholds



The simulation highlights the operational compromise between screening sensitivity and administrative burden. At a threshold of 0.20, almost 2.03 million households would be identified for potential intervention, exhibiting a targeting accuracy of 58.4%. Raising the criterion to 0.40 diminishes the workload to around 1.02 million families, concurrently enhancing precision to 69.0%. At a threshold of 0.60, the burden decreases to around 0.40 million homes, with precision increasing to 76.6%.

The results indicate that threshold selection should align with the administrative capabilities of housing organizations. Reduced thresholds detect a greater number of potentially at-risk families but may create workloads that beyond operational capabilities. Elevated thresholds diminish caseload number and enhance targeting accuracy, although they concurrently restrict program accessibility.

Therefore, prediction thresholds should not be regarded as immutable statistical characteristics. They should be regarded as policy instruments that can be adjusted according to available resources, personnel capabilities, and program goals, enabling housing authorities to reconcile coverage with operational practicality.

4.6 Strategy Comparison: Machine Learning vs Traditional Policy

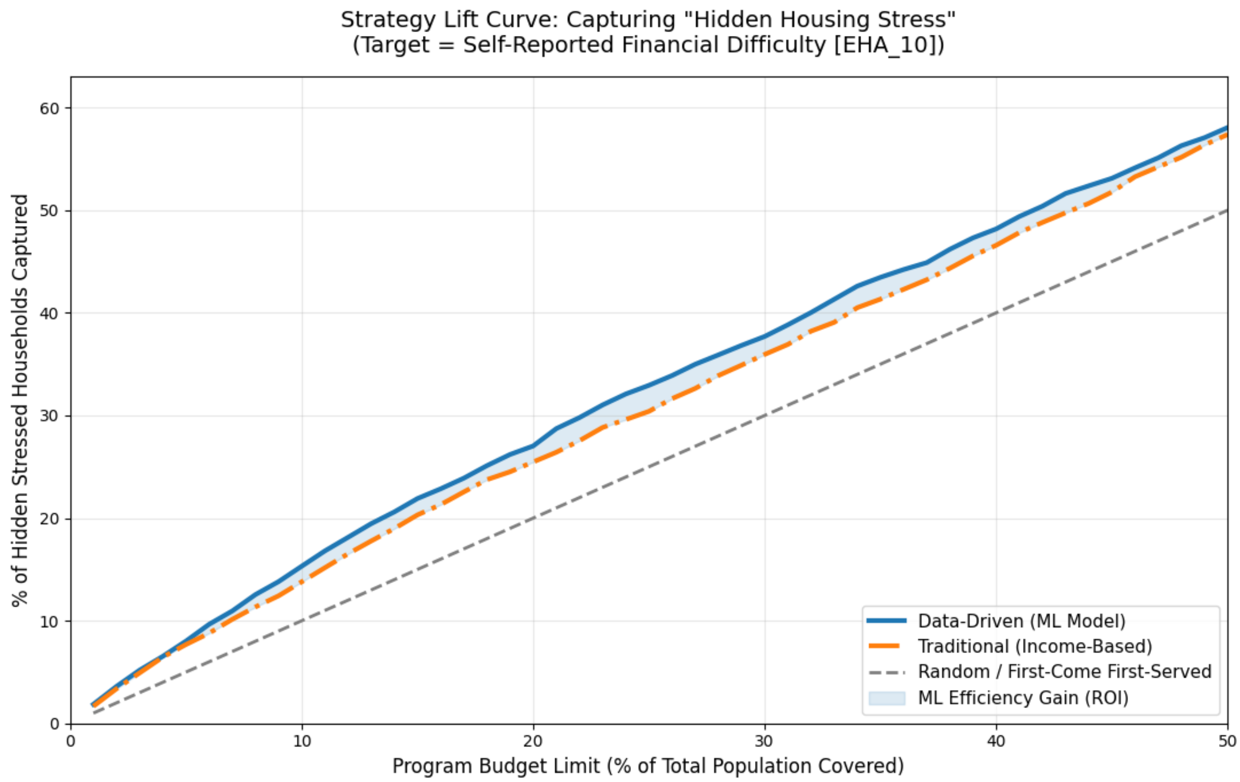
A simulation was run to compare strategies and evaluate the operational utility of the prediction model. The objective was to assess if risk-based targeting surpasses conventional housing policy methods within fixed budget limitations.

Three allocation procedures were evaluated:

1. Random selection (FCFS baseline)
2. Traditional income-based targeting
3. Machine learning risk-based targeting

Each technique was simulated within uniform budget constraints to assess its efficacy in identifying households with housing difficulty.

Figure 12 Strategy Performance Comparison Under Budget Constraint



When the objective was official Core Housing Need, income-based targeting yielded comparable results to the machine learning methodology. This is anticipated as the formal definition of housing necessity is significantly linked to income.

Nevertheless, when the focus transitioned to self-reported financial hardship ("Hidden Stress"), the machine learning approach exhibited superior performance. With an appropriate program budget of roughly 24% coverage, the machine learning approach identified 32.1% of concealed stress households, whereas income-based targeting identified 29.6%, resulting in an operational advantage of approximately 2.5%.

Despite the seemingly minor numerical change, within the context of the Canadian housing system, this signifies tens of thousands of additional families accurately recognized. The findings indicate that housing stress is multifaceted and cannot be entirely represented by income limits alone.

4.7 Interpretable Policy Decision Matrix

To translate predictive outputs into actionable policy guidance, a policy decision matrix was developed using the model's predicted risk scores. Instead of considering risk probabilities as mere abstract figures, households were categorized into four operational levels according to their standing in the national risk distribution.

The population was divided into four policy tiers using predicted risk percentiles:

1. Tier 1 – Critical Risk: Top 10% of predicted probabilities
2. Tier 2 – High Risk: 70th–90th percentiles
3. Tier 3 – Moderate Risk: 40th–70th percentiles
4. Tier 4 – Low Risk: Bottom 40%

For each tier, weighted household characteristics were calculated using survey weights to estimate national population totals and demographic patterns.

Table 4 Policy Decision Matrix Based on Predicted Housing Stress Risk

Action Tier	Est. Households (M)	Avg Income	% Renters	Hidden Stress Rate	Prescribed Policy Intervention
Tier 1: Critical (Top 10%)	0.59M	$22,949	80.5%	55.9%	Emergency Rent Relief / Direct Cash Subsidy

Tier 2: High Risk (Next 20%)	1.48M	$35,254	57.8%	42.8%	Targeted Tax Credits / Rent Control Protections
Tier 3: Moderate (Next 30%)	3.92M	$58,495	47.5%	33.4%	Affordable Housing Supply / First-Time Buyer Incentives
Tier 4: Low Risk (Bottom 40%)	8.61M	$163,757	21.2%	25.0%	No Intervention / Standard Zoning Policies

The resulting framework converts model predictions into explicit policy actions.

1. Tier 1 (Critical Risk): Approximately 590,000 households, with an average income of $22,949, a predominant renter demographic (80.5%), and significant financial difficulties (55.9%). These households necessitate immediate housing support, encompassing emergency rent assistance or direct financial subsidies.
2. Tier 2 (High Risk): Approximately 1.48 million families with an average income of $35,254 and a rental proportion of 57.8%. Policy solutions may encompass specific tax credits, rent stabilization initiatives, or housing subsidies.
3. Tier 3 (Moderate Risk): Approximately 3.92 million households, typically middle-income, facing affordability challenges. Policy interventions at this level concentrate on market stabilization techniques, such as increasing the supply of affordable housing and assisting first-time homebuyers.

4. Tier 4 (Low Risk): Approximately 8.61 million households exhibit considerable financial stability and a reduced renter proportion (21.2%). These households often necessitate minimal direct involvement, as conventional housing market policies are enough.

This matrix serves as an effective decision-support instrument by converting predictive risk ratings into distinct operational categories, allowing housing agencies to synchronize intervention tactics with the intensity of housing stress within the population.

Chapter V: Data Visualization and Interactive Decision Support

5.1 Visualization Strategy

To improve interpretability and facilitate policy-oriented research, the findings of this study were presented using interactive visualization platforms instead of exclusively utilizing static charts. Two complementing instruments were created: a Streamlit application for interactive model exploration and policy simulation, and a Power BI dashboard for national-level monitoring and visual representation of housing stress patterns.

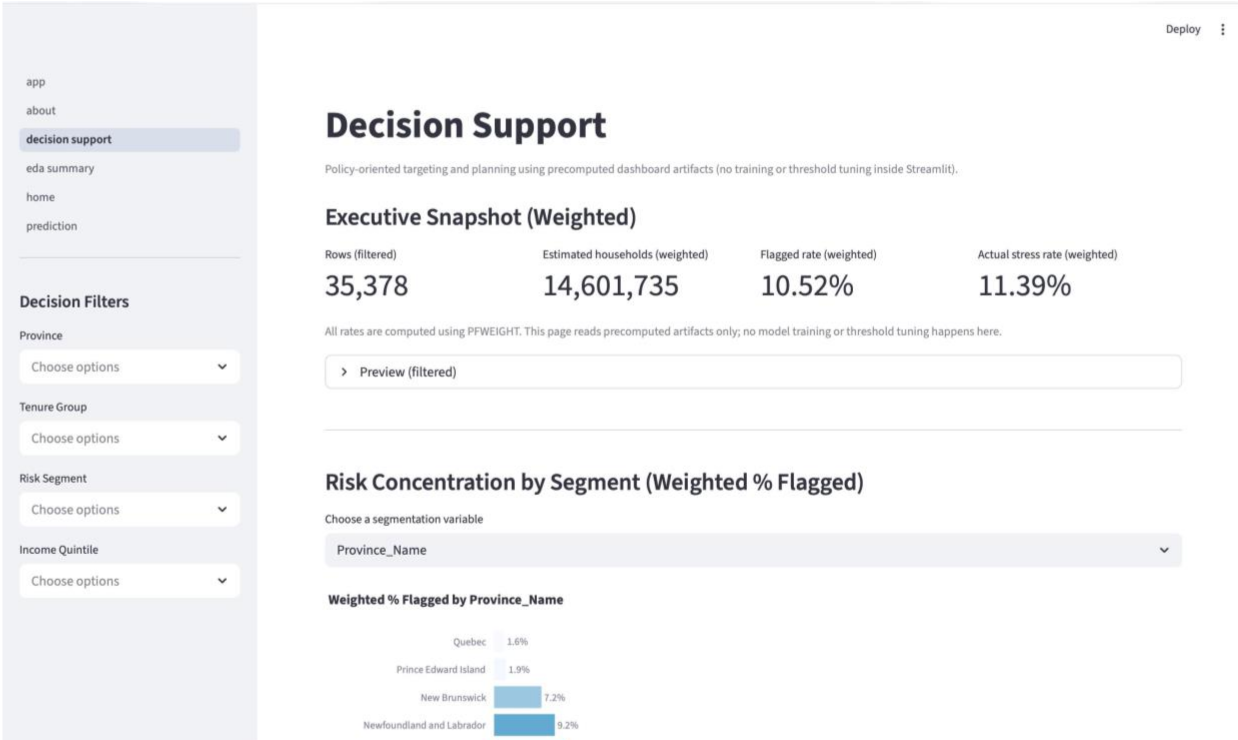
These visualization tools enable users to examine the analytical results produced in prior chapters, encompassing housing stress distributions, model projections, and resource allocation scenarios. This chapter emphasizes the most pertinent visualizations for decision-making, while supplementary diagnostic charts are included in Appendix B.

5.2 Streamlit Decision Support Dashboard

A decision-support interface was created with Streamlit to enable policymakers and analysts to examine predictive model outputs and simulate housing policy targeting options.

Executive Decision Overview

Figure 13 Streamlit Executive Snapshot Dashboard



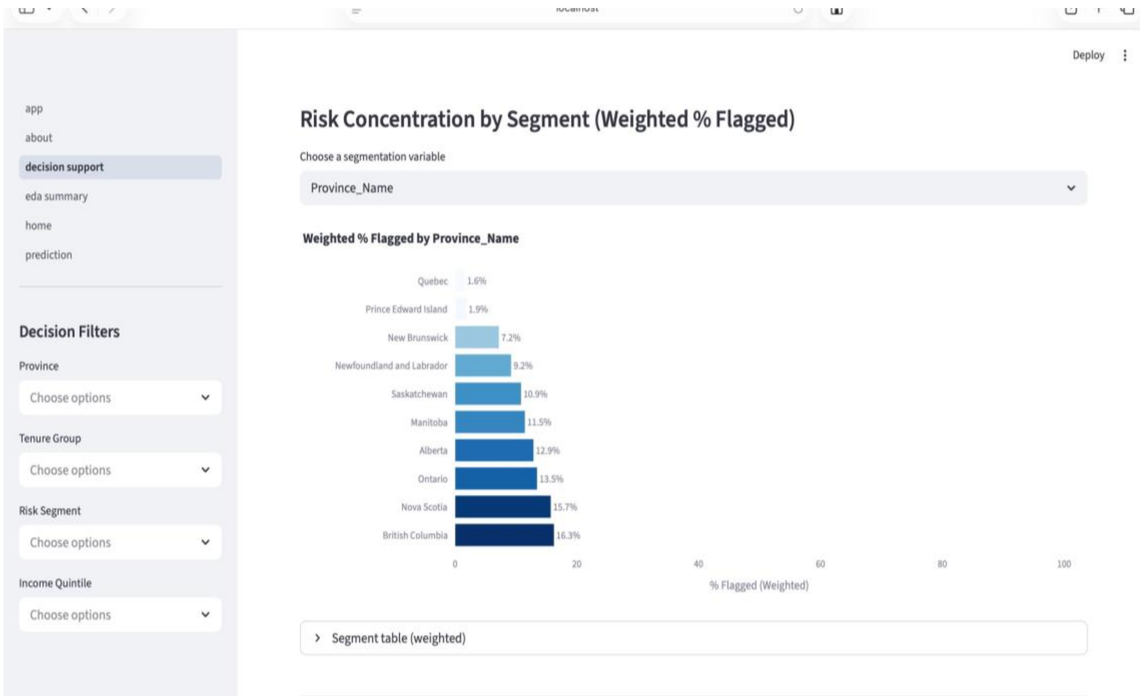
The dashboard indicates that roughly 11.39% of Canadian households face housing affordability challenges, whereas the predictive model identifies 10.52% of households as high risk according to the operational screening criterion.

The strong correlation between the observed stress rate and the anticipated high-risk share suggests that the model effectively identifies a significant number of at-risk homes while keeping the screening population within reasonable limits.

Risk Concentration by Household Characteristics

The Streamlit interface enables users to analyze the variation of expected housing stress across various structural household features.

Figure 14 Streamlit Risk Concentration by Province



The provincial segmentation perspective demonstrates that housing stress is inequitably spread throughout Canada. Some provinces exhibit elevated anticipated stress levels, indicating regional disparities in home affordability challenges.

Supplementary segmentation perspectives are accessible in the dashboard for:

1. Tenure Group
2. Income Quantile
3. Household Composition
4. Risk Categories

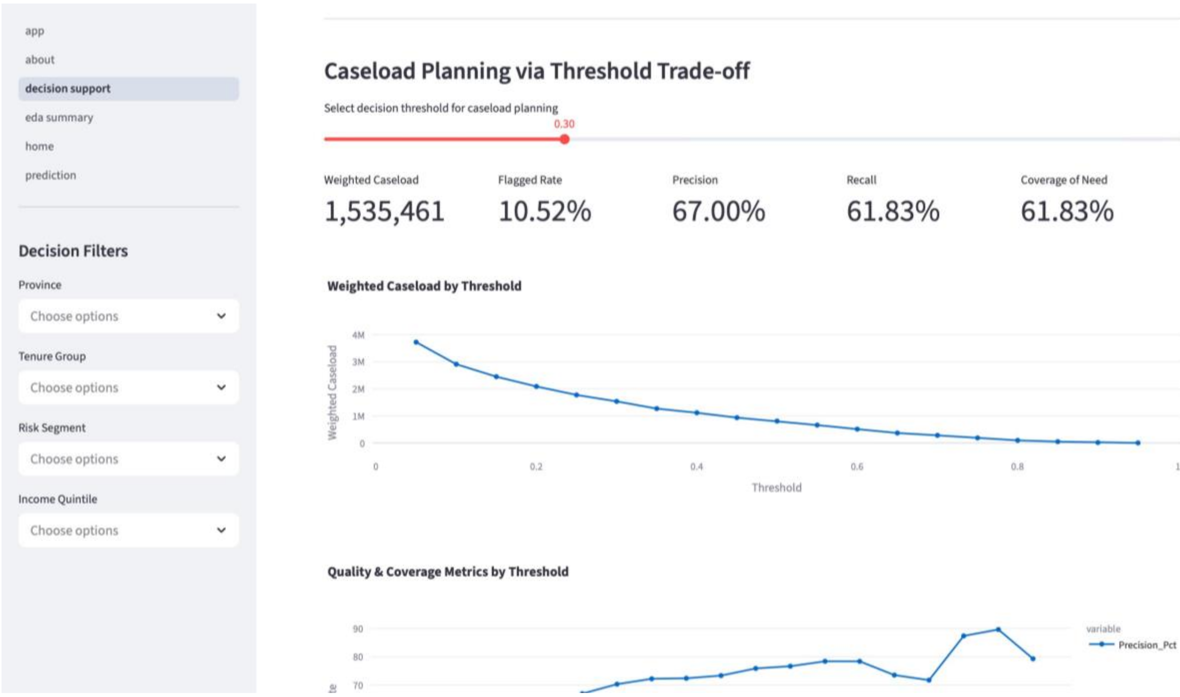
These interactive infographics enable users to examine the variations in housing stress among different demographic groups. Tenure segmentation consistently indicates that renters have significantly greater housing stress than homeowners, corroborating findings observed in the descriptive study.

Additional segmentation charts are available in Appendix B (Streamlit Visualization Screens)**.**

Threshold Planning and Weighted Caseload Simulation

The Streamlit application prominently contains a caseload planning module that illustrates the trade-offs among model precision, recall, and administrative workload.

Figure 15 Streamlit Caseload Planning Threshold Trade-off chart



The visualization illustrates how modifications to the categorization threshold affect the quantity of homes identified for potential policy intervention. At the designated operational threshold of 0.30, the model identifies roughly 1.53 million households, constituting 10.52% of the weighted household population.

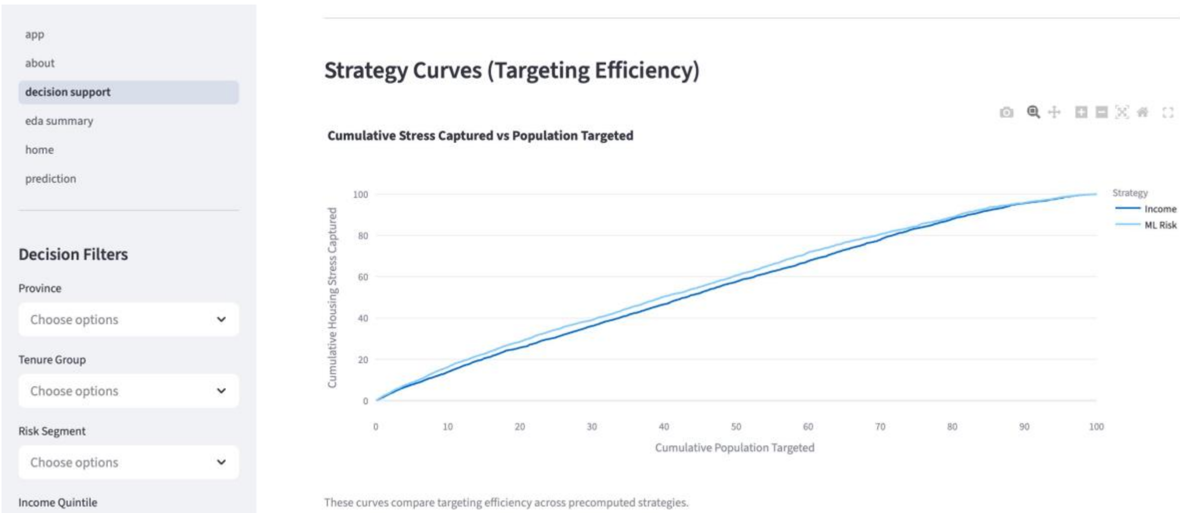
This visualization offers policymakers a pragmatic instrument for assessing the correlation between screening sensitivity and administrative capability.

Supplementary threshold diagnostic charts (precision, recall, and coverage curves) are provided in Appendix B.

Strategy Comparison: Targeting Efficiency

The Streamlit interface features a graphic that compares various policy targeting tactics.

Figure 16 Strategy Curves (Targeting Efficiency)



The strategy curves evaluate the efficacy of conventional income-based targeting against machine-learning risk targeting. The findings suggest that risk-based targeting more effectively identifies housing stress throughout the population distribution.

Despite the minor improvement, the machine-learning approach reliably detects a little greater percentage of stressed households at comparable targeting coverage levels.

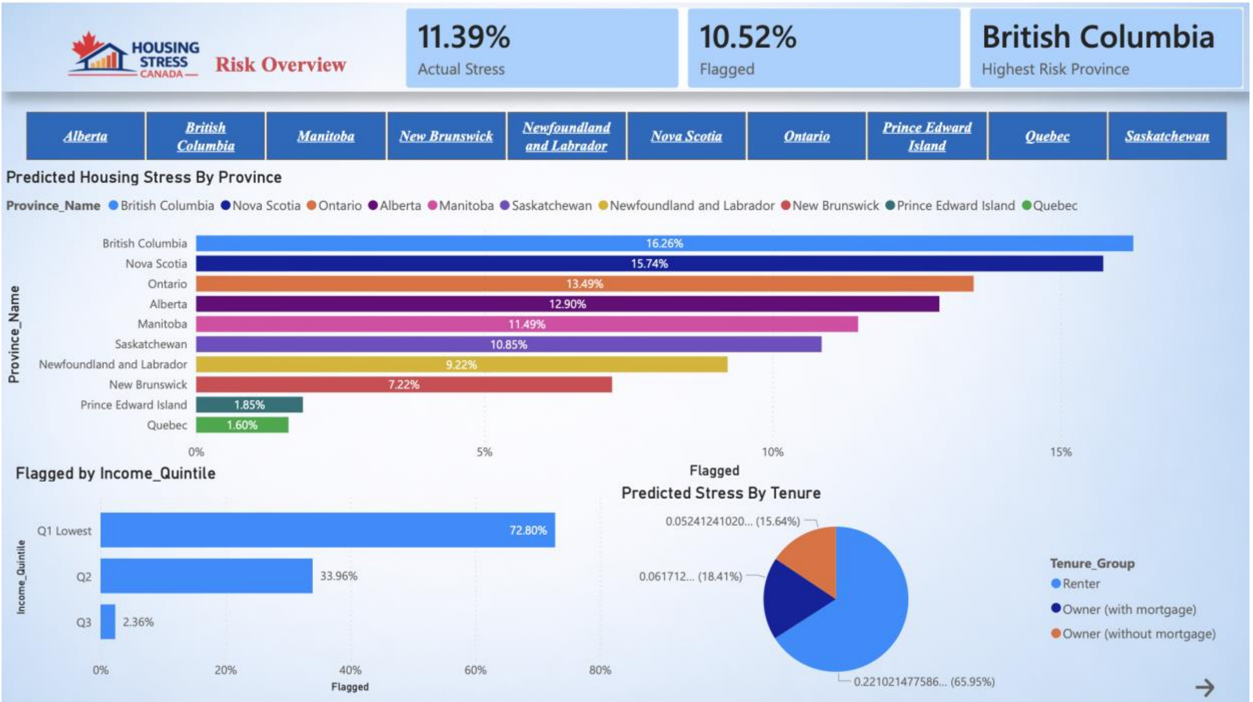
5.3 Power BI Housing Stress Dashboard

A Power BI dashboard was created to convey critical findings inside an organized visual monitoring environment, in addition to the interactive analytical interface.

The dashboard consolidates national housing stress metrics and offers visual comparisons among provinces, tenure types, and income levels.

Overview of National Housing Stress

Figure 17 Power BI Risk Overview Dashboard

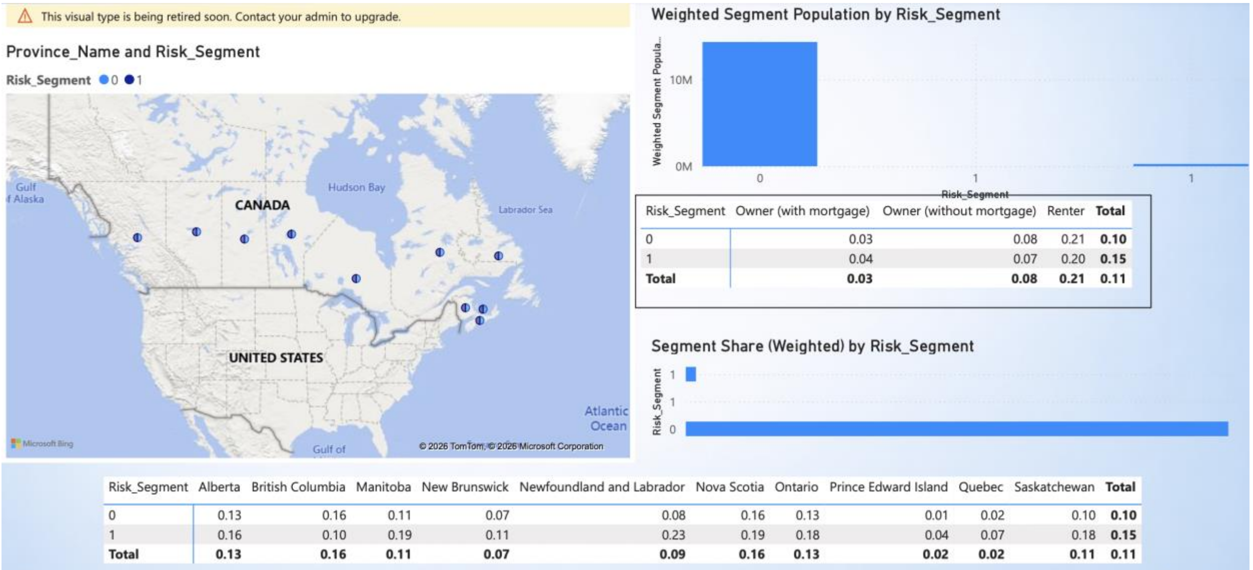


The overview panel presents essential national metrics, such as the weighted prevalence of housing stress (11.39%) and the proportion of households identified by the predictive model (10.52%).

The dashboard emphasizes regional disparities in anticipated housing stress rates, designating British Columbia as the province with the highest projected housing stress.

Provincial Housing Stress

Figure 18 Power BI Forecasted Housing Distress by Province



The province comparison graphic indicates significant variation in projected housing stress rates throughout Canada. British Columbia demonstrates the greatest anticipated stress rate at roughly 16.26%, succeeded by Nova Scotia and Ontario.

These patterns illustrate the disparate distribution of home affordability challenges within Canadian housing markets.

Housing Stress by Tenure and Income

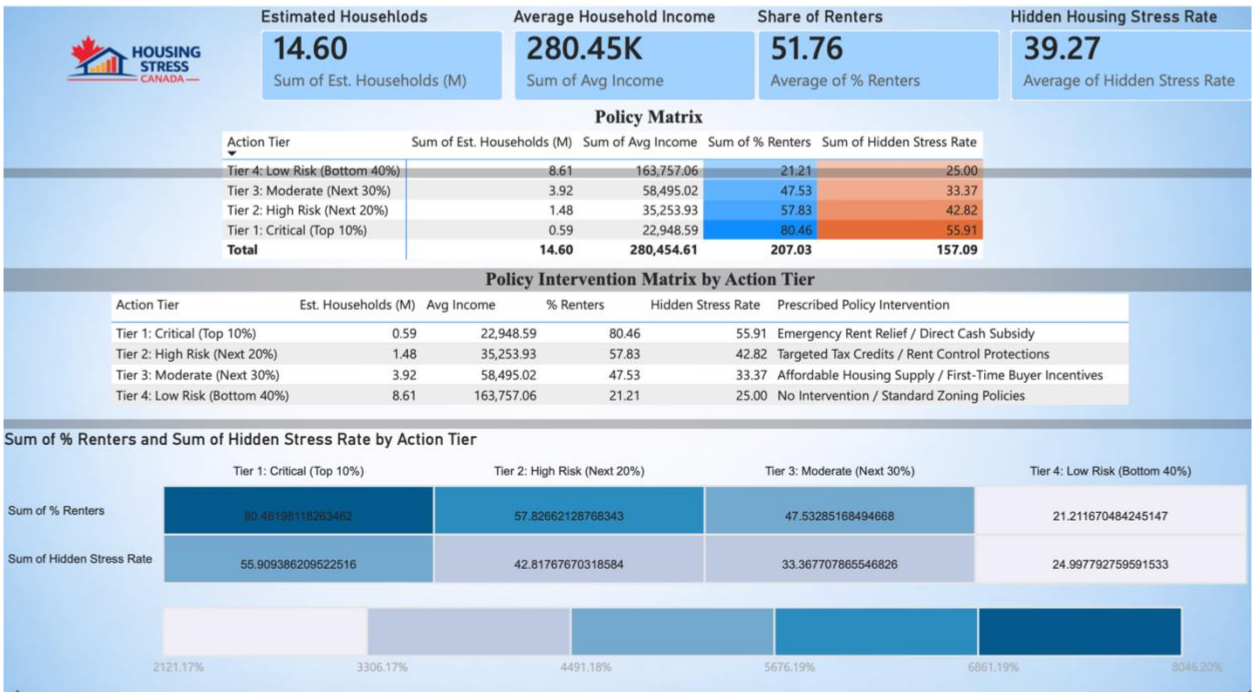
The Visualization shown in Figure 18 above indicates that roughly 65.95% of families identified as at-risk are renters, whereas homeowners constitute a lesser proportion of anticipated stress situations.

The income distribution picture further substantiates that housing stress is predominantly experienced by lower-income households. The lowest income quintile constitutes nearly 70% of anticipated housing stress cases, corroborating the income gradient revealed in the exploratory research.

5.4 Policy Decision Matrix Visualization

The final dashboard element illustrates the policy intervention matrix, which converts anticipated risk scores into implementable policy levels.

Figure 19 Power BI Policy Intervention Matrix



The matrix categorizes the population into four policy levels according to anticipated housing stress risk.

The visualization indicates that high-risk households are a relatively small segment of the population yet demonstrate markedly greater renter concentration and financial vulnerability, underscoring the necessity for focused governmental actions.

5.5 Additional Visualization

Additional diagnostic visualizations generated throughout the analysis are included in Appendix B, comprising:

1. Model confusion Matrix
2. PSTIR validation gradients charts
3. Additional segmentation breakdowns
4. Supplementary threshold trade-off charts

The additional visuals strengthen the analytical framework's robustness, enabling the main report to concentrate on the most significant policy results.

Chapter VI: Findings and Discussion

6.1 Key Empirical Findings

The analysis reveals multiple persistent patterns in the distribution and prediction of housing affordability stress among Canadian households. Initially, housing stress is not uniformly distributed throughout tenure types. Renters demonstrate significantly higher levels of housing stress as compared to homeowners, suggesting that engagement with the rental market is a primary structural driver contributing to affordability issues.

A strong income inequality is seen. Housing stress is predominantly prevalent among lower-income households, with the lowest income quintile representing an excessively significant number of stressed households. This pattern is uniform in both descriptive and predictive analyses, underscoring the pivotal role of economic capability in influencing housing affordability outcomes.

Third, regional disparities are apparent among provinces. Housing stress levels significantly vary by geographic location, indicating differences in housing market factors, such as cost pressures and supply limitations. Provinces like British Columbia and Nova Scotia demonstrate elevated predicted stress rates compared to other areas.

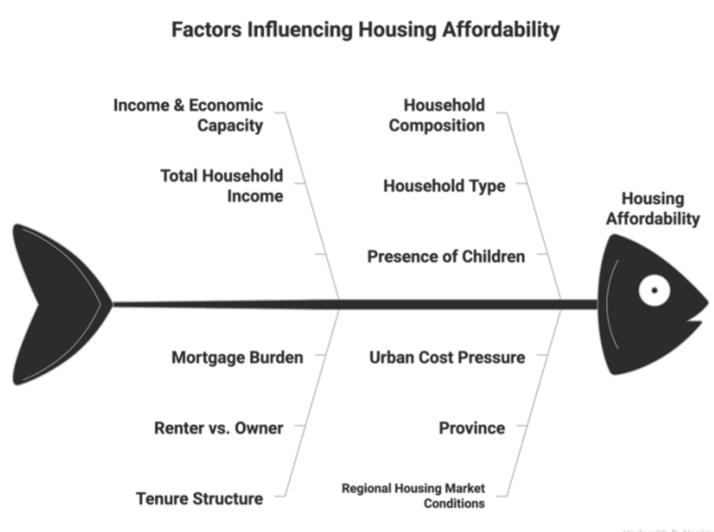
The logistic regression model exhibits robust efficacy as a screening method from a predictive standpoint. The model attains a ROC-AUC of 0.9188, signifying a strong capacity to differentiate between households undergoing housing difficulty and those not in need. At the chosen operational threshold of 0.30, the model achieves a recall rate of 0.80, thereby detecting a significant number of households facing housing hardship, while sustaining a moderate level of precision.

The weighted population estimates reveal that around 11.39% of Canadian families face home affordability stress, but the predictive model identifies roughly 10.52% of households as high risk based on the chosen threshold. This close alignment indicates that the model effectively balances the identification of vulnerable families with the maintenance of a manageable screening population for policy implementation.

Collectively, these findings affirm that housing affordability stress in Canada is organized along distinct economic, tenure, and geographic dimensions, and that these patterns can be effectively utilized to facilitate household-level risk assessment.

The drivers of housing affordability stress identified in this study can be summarized through a conceptual framework that integrates economic, demographic, and structural housing factors.

Figure 20 Conceptual Framework of Housing Affordability Stress Drivers



The framework highlights how income capacity, tenure structure, household composition, and regional housing market conditions jointly contribute to housing affordability stress. These dimensions align with the empirical findings presented in earlier chapters, reinforcing that housing stress is shaped by multiple interacting factors rather than a single determinant.

6.2 Hidden Housing Stress

The analysis reveals that official measures of housing affordability stress do not fully capture the distribution of financial pressure across households.

Key observations:

1. Official (weighted) Core Housing Need: 11.39%
2. Model-identified high-risk households: 10.52%

Although these values are closely correlated, further study indicates that housing stress transcends the previously recognized vulnerable categories.

Key Insights:

1. Housing stress is not confined to the lowest income group.
2. A significant share of moderate-income households (Tier 3) exhibit affordability pressure
3. These households are often not captured by strict income-based eligibility criteria.

Evidence from Policy Segmentation:

1. Tier 2 (High Risk): 1.48M households
2. Tier 3 (Moderate Risk): 3.92M households

Collectively, these categories constitute a significant segment of households facing non-extreme yet persistent affordability challenges.

Interpretation:

1. Traditional policy frameworks tend to focus on extreme need (Tier 1).

2. However, the model identifies a broader group of households experiencing "hidden housing stress", particularly among:
    a. renters outside the lowest income bracket.
    b. households facing high-cost burdens relative to income

Implications:

1. Housing affordability stress in Canada is gradient based, not binary.
2. Relying solely on income thresholds risks under-identifying at-risk households.
3. Data-driven risk scoring provides a more nuanced and inclusive identification framework.

6.3 Policy Implications

The findings emphasize the necessity for a transition from standardized housing policy methods to focused, risk-oriented intervention measures.

Key Implications:

1. Housing stress exists on a spectrum rather than in a binary format, defined by varying levels of risk.
2. A single eligibility rule (e.g., income threshold) is insufficient.
3. Policy formulation must account for diverse needs and resource limitations.

Tier-Based Policy Framework

The policy decision matrix offers a systematic framework for intervention.

1. Tier 1 (Critical Risk – 0.59M households): Immediate assistance necessary immediate rental assistance, direct financial subsidies.

2. Tier 2 (High Risk – 1.48 million households): Eliminate escalation through targeted tax incentives and rent protections.
3. Tier 3 (Moderate Risk – 3.92 million households): Emphasis on stabilization provision of affordable housing and buyer support programs.
4. Tier 4 (Low Risk – 8.61 million households): Minimal involvement standard housing and zoning regulations.

Operational Insight:

1. The model supports prioritization under limited budgets.
2. At a threshold of 0.30, roughly 10.52% of households are identified, nearly corresponding to the national stress rate of 11.
3. This facilitates effective targeting without weighing down administrative resources.

Strategic Insight

The housing policy must evolve from:

1. Broad, uniform program to targeted, data-driven allocation strategies
2. Risk-based targeting improves:
    a. Efficiency (who gets help).
    b. Equity (who needs help).

6.4 Contribution of the Study

This study contributes to the housing affordability literature and policy practice by creating a household-level, data-driven decision-support system.

Key Contributions:

1. Microdata-driven analysis: Utilizes CHS 2022 household-level data to move beyond aggregate housing indicators.

2. Integrated analytics pipeline: Merges:
    a. Descriptive analysis
    b. Predictive modeling
    c. Prescriptive optimization
3. Interpretable risk modeling uses logistic regression to generate transparent and policy-relevant risk scores, supported by a Random Forest benchmark.
4. Operational decision-support tools:

    Translate model outputs into:
    a. Risk Segmentation.
    b. Decision rules.
    c. Policy tier Metrix.
5. Resource allocation farmwork:

    Exhibits the prioritization of housing support within budgetary limitations through threshold optimization and targeting simulations.

Overall Contribution

The study transitions housing affordability analysis from descriptive reporting to actionable decision support, thereby allowing policymakers to more effectively identify, prioritize, and address housing stress at the household level.

6.5 Limitations

1. Cross-sectional data:

    The CHS 2022 dataset provides a snapshot of housing conditions at a specific moment and does not indicate changes in housing stress over time.
2. Measurement constraints:

Core Housing Need (PCHN) may not fully capture all forms of hidden or self-reported housing stress, especially within moderate-income households.

3. Geographical limitation:

   The use of the PUMF (Public Use Microdata File) confines study to larger regions and restricts comprehensive local or municipal-level insights.

4. Non-casual modeling:

   The predictive framework is designed for risk identification rather than causal inference. While the model captures statistically meaningful associations between household characteristics and housing stress, it does not establish causal relationships.

Chapter VII: Conclusion and Recommendations

7.1 Conclusion

This study analyzed housing affordability stress in Canada utilizing household-level data from the 2022 Canadian Housing Survey (CHS). The project established a data-driven framework for identifying and prioritizing households at risk of housing stress by integrating weighted descriptive analysis, interpretable predictive modeling, and prescriptive analytics.

The results indicate that housing affordability stress is organized according to essential factors, notably income, tenure, and geographic location. Renters and low-income households have the greatest levels of stress, with geographical disparities further influencing the distribution of housing challenges throughout Canada.

Importantly the analysis indicates that housing stress exists on a continuum of risk rather than as a binary condition, and that predictive modeling may accurately identify households at elevated risk. The use of risk-based segmentation, threshold optimization, and policy tiering demonstrates how analytical techniques can enhance the efficient and targeted distribution of scarce housing resources.

7.2 Recommendation

Based on the findings, several policy and analytical recommendations emerge:

1. Implement risk-based targeting frameworks:

   Housing programs ought to transcend singular eligibility criteria and integrate predictive risk assessment to more effectively identify households requiring assistance.

2. Implement tiered policy interventions:

   Different levels of housing stress require differentiated responses, ranging from emergency aid for high-risk households to market stabilization policies for moderate-risk groups.

3. Align program thresholds with administrative capacity:

   Decision thresholds should be treated as policy parameters, allowing agencies to balance coverage of need with operational feasibility.

4. Enhance data integration for housing policy:

   Future initiatives should amalgamate survey data with administrative and market data to improve targeted accuracy and geographic accuracy.

5. Support ongoing analytical development:

   Housing policy design can benefit from continued use of interpretable machine learning and decision-support tools to enhance transparency and effectiveness.

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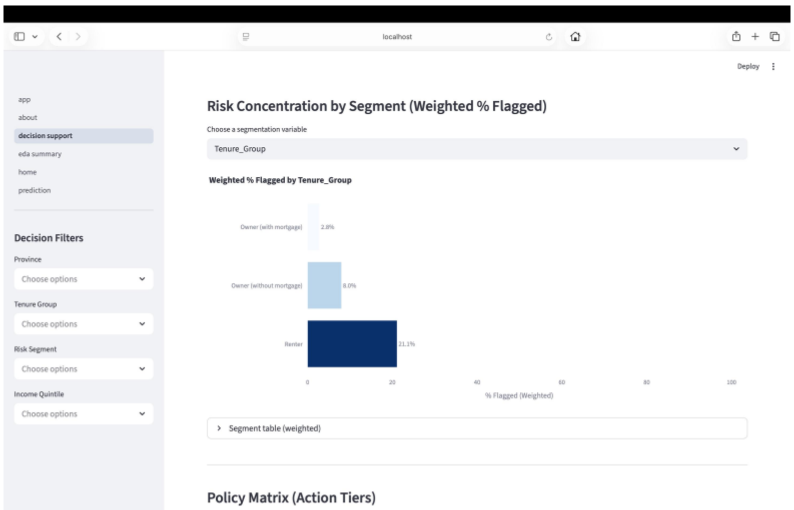
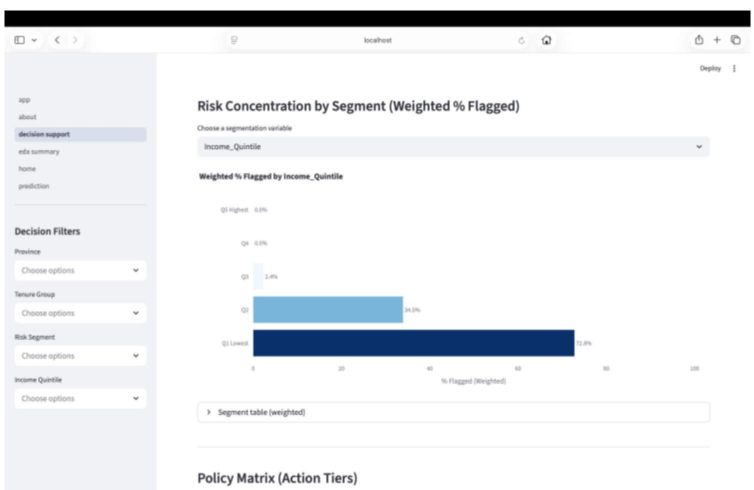
Appendix

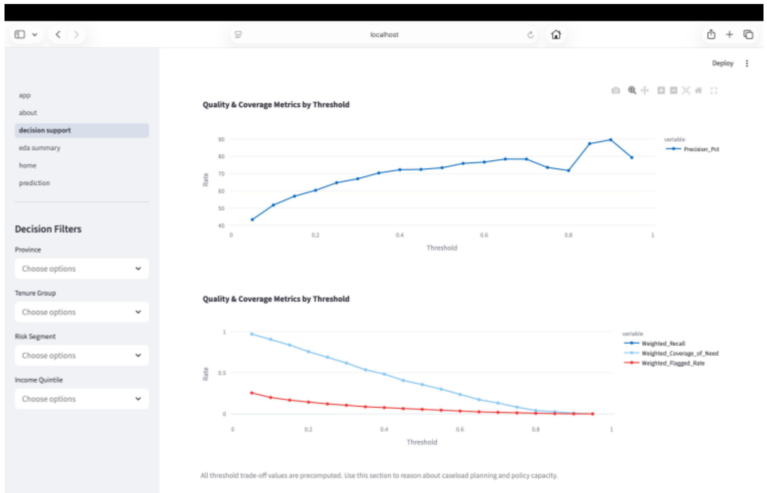
Appendix A

https://www150.statcan.gc.ca/n1/en/catalogue/46250001

Appendix B



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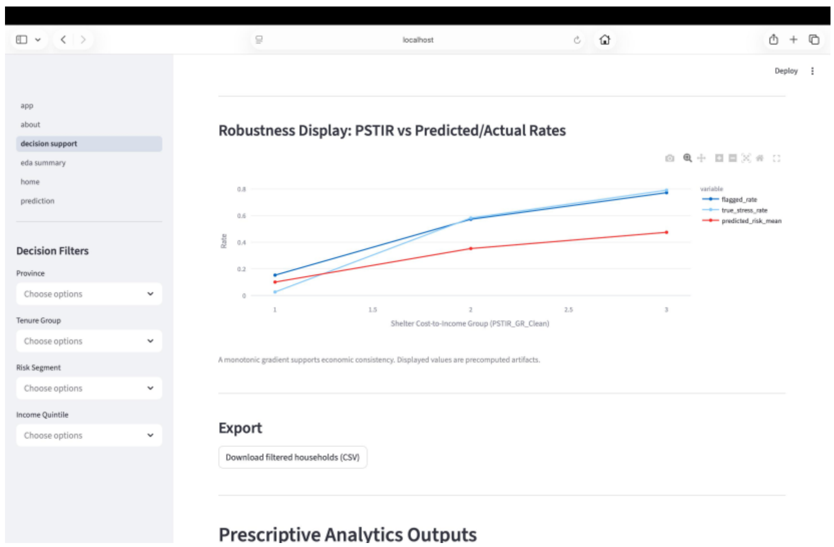
All threshold trade-off values are precomputed. Use this section to reason about caseload planning and policy capacity.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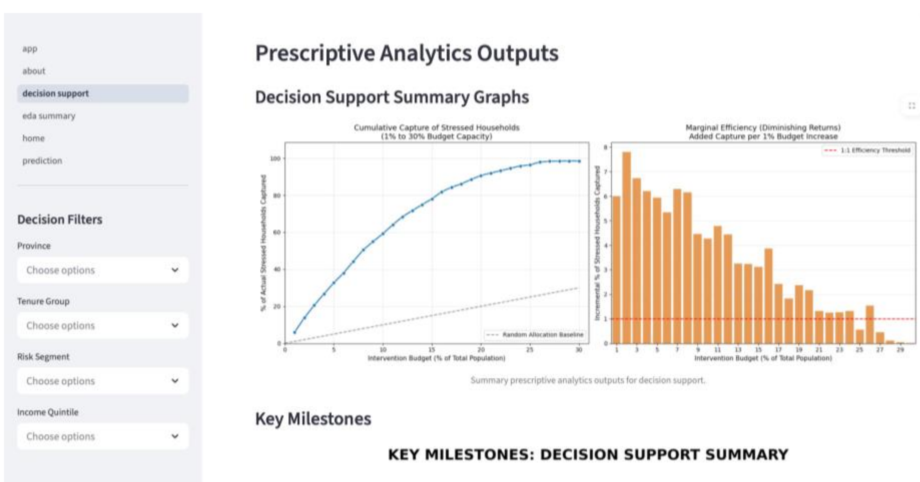
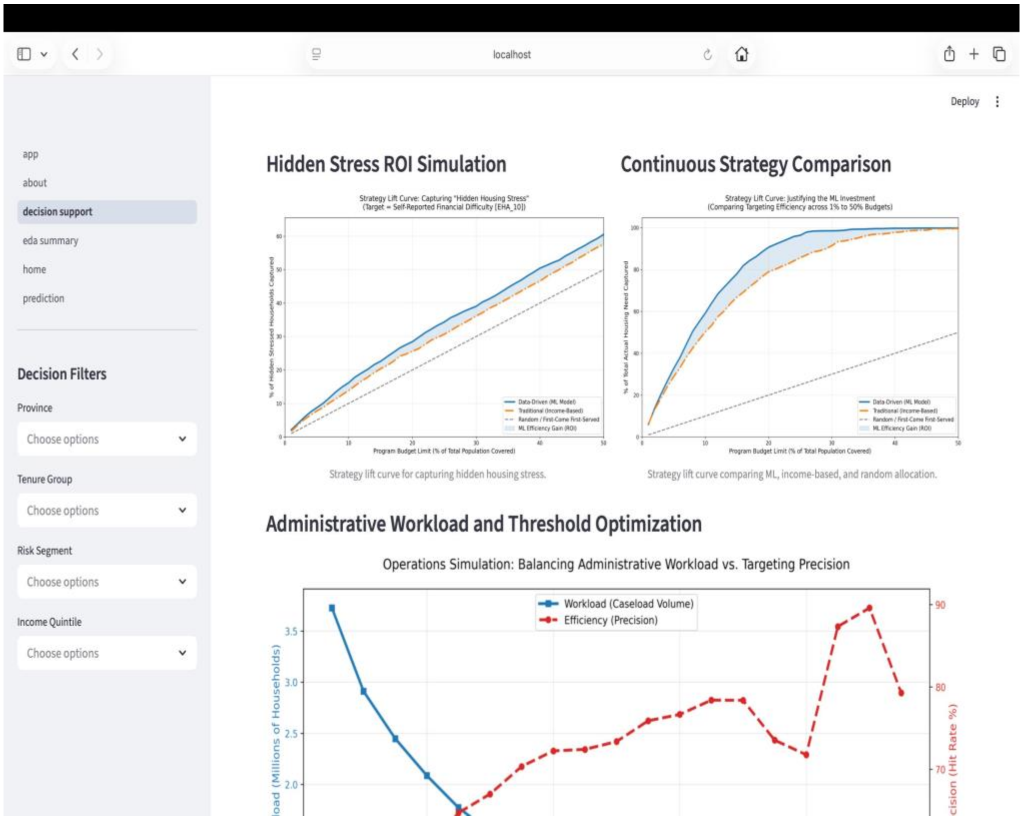
Rate
Shelter Cost-to-Income Group (PSTIR_GR_Clean)
A monotonic gradient supports economic consistency. Displayed values are precomputed artifacts.
Export
Download filtered households (CSV)
Prescriptive Analytics Outputs


app
about
decision support
eda summary
home
prediction
Decision Filters
Province
Choose options
Tenure Group
Choose options
Risk Segment
Choose options
Income Quintile
Choose options
Prescriptive Analytics Outputs
Decision Support Summary Graphs
Cumulative Capture of Stressed Households
(1% to 30% Budget Capacity)
% of Actual Stressed Households Captured
Random Allocation Baseline
Intervention Budget (% of Total Population)
Marginal Efficiency (Diminishing Returns)
Added Capture per 1% Budget Increase
1:1 Efficiency Threshold
Incremental % of Stressed Households Captured
Intervention Budget (% of Total Population)
Summary prescriptive analytics outputs for decision support.
Key Milestones
KEY MILESTONES: DECISION SUPPORT SUMMARY


localhost
Deploy
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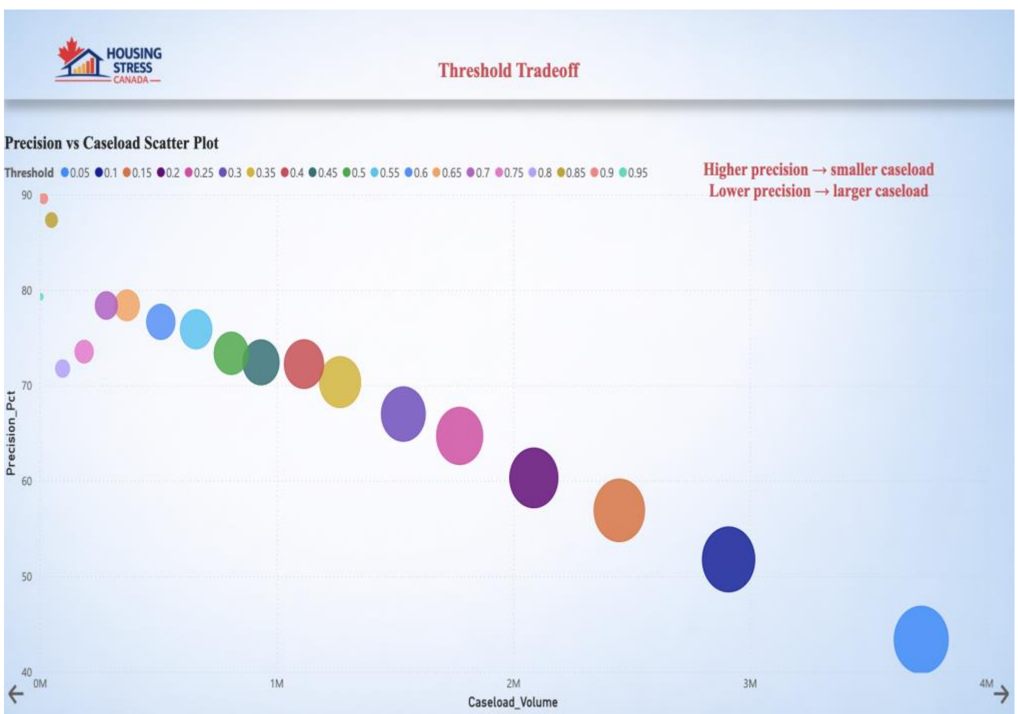
about
decision support
eda summary
home
prediction
Decision Filters
Province
Choose options
Tenure Group
Choose options
Risk Segment
Choose options
Income Quintile
Choose options
Hidden Stress ROI Simulation
Continuous Strategy Comparison
Strategy lift curve for capturing hidden housing stress.
Strategy lift curve comparing ML, income-based, and random allocation.
Administrative Workload and Threshold Optimization
Operations Simulation: Balancing Administrative Workload vs. Targeting Precision
Workload (Caseload Volume)
Efficiency (Precision)


HOUSING
STRESS
CANADA
Threshold Tradeoff
Precision vs Caseload Scatter Plot
Threshold 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8 0.85 0.9 0.95
Higher precision → smaller caseload
Lower precision → larger caseload
Precision_Pct
Caseload_Volume
0M
1M
2M
3M
4M